INCH-POUND

MIL-M-28787/517
5 May 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

SHIFT MATRIX AND GENERAL REGISTER MODULE, KEY CODE CWE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic shift matrix and general register module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 3.077 failures per million operating hours for class I or 8.168 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be CWE.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
A2	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	---------Module shall pass all remaining tests---------						
D1	Ibd	V	S	0.0	3.1	0.0	3.1	0.0	3.1	mA
D2,D3	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D4-D8	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	mA
D10,D11	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D12	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D13	Ibd	V	S	0.0	2.1	0.0	2.1	0.0	2.1	mA
D14	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D15-D24	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D25	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D26-D41	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D42-D47	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D48	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D49-D53	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
E1-E11	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E12	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E13-E15	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E16	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E17-E19	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E20	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E21-E23	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E24	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E25-E27	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E28	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E29-E31	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E32	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E33-E35	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E36	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 3°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
E37-E39	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E40	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E41	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E42	Ios	VI	S	-100.0	-15.0	-100.0	-15.0	-100.0	-15.0	mA
E43-E50	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E51,E52	Ios	VI	S	-90.0	-20.0	-90.0	-20.0	-90.0	-20.0	mA
E53-E55	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E56	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E57-E59	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E60	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E61-E63	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E64	Ios	VI	S	-220.0	-30.0	-220.0	-90.0	-220.0	-30.0	mA
E65-E67	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E68	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E69-E71	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E72	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E73-E75	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E76	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E77-E79	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E80	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E81-E83	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E84	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
F1	Ith	VII	X	-6.6	0.17	-6.6	0.17	-6.6	0.17	mA
F2,F3	Ith	VII	X	-0.8	0.02	-0.8	0.02	-0.8	0.02	mA
F4-F11	Ith	VII	X	-2.85	0.11	-2.85	0.11	-2.85	0.11	mA
F12-F17	Ith	VII	X	-0.85	0.045	-0.85	0.045	-0.85	0.045	mA
F18,F19	Ith	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F20	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F21	Ith	VII	X	-4.8	0.12	-4.8	0.12	-4.8	0.12	mA
F22	Ith	VII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
F23	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F24-F28	Ith	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F29-F32	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 50 ∓ 5°C		Class II 4/ -55 ± 5°C 85 ∓ 5°C		Units
				Min	Max	Min	Max	Min	Max	
F33	Ilh	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F34-F37	Ilh	VII	X	-2.05	0.11	-2.05	0.11	-2.05	0.11	mA
F38-F41	Ilh	VII	X	-2.65	0.13	-2.65	0.13	-2.65	0.13	mA
F42-F49	Ilh	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F50-F57	Ilh	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F58-F63	Ilh	VII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
F64	Ilh	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F65-F69	Ilh	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G1	Iil	VIII	X	-6.6	0.17	-6.6	0.17	-6.6	0.17	mA
G2,G3	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G4-G11	Iil	VIII	X	-2.05	0.11	-2.05	0.11	-2.05	0.11	mA
G12-G17	Iil	VIII	X	-0.85	0.045	-0.85	0.045	-0.85	0.045	mA
G18,G19	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G20	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G21	Iil	VIII	X	-4.6	0.12	-4.6	0.12	-4.6	0.12	mA
G22	Iil	VIII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
G23	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G24-G28	Iil	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G29-G32	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G33	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G34-G37	Iil	VIII	X	-2.05	0.11	-2.05	0.11	-2.05	0.11	mA
G38-G41	Iil	VIII	X	-2.65	0.13	-2.65	0.13	-2.65	0.13	mA
G42-G49	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G50-G57	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G58-G63	Iil	VIII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
G64	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G65-G69	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
H1-H84	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J85	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
K1-K16	Iozh	XI	X	-0.17	0.17	-0.17	0.17	-0.17	0.17	mA
K17-K32	Iozl	XI	X	-0.17	0.17	-0.17	0.17	-0.17	0.17	mA
L1-L13	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L14	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L15	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L16-L20	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L21-L25	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L26-L33	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L34-L41	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L42-L47	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L48-L53	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M1	PSFC	XIII	S	4.9	9.1	4.9	9.1	4.9	9.1	uF
M2	PSFC	XIII	S	6.3	11.7	6.3	11.7	6.3	11.7	uF
N	Ioo	XIV	X	900	1920	900	1920	900	1920	mA
P1,P2	Functional	XV	X	----------		See table XV				
Q1,Q2	Dynamic	XVI	S	----------		See table XVI				
Q3,Q4	Dynamic	XVI	Q	----------		See table XVI				
R1	Ci/o	XVIII	Q	0	125	0	125	0	125	pF
R2,R3	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R4-R11	Ci/o	XVIII	Q	4	100	4	100	4	100	pF
R12	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R13	Ci/o	XVIII	Q	6	100	6	100	6	100	pF
R14	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R15	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R16-R18	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R19-R24	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R25	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R26-R47	Ci/o	XVIII	Q	2	50	2	50	2	50	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R48	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R49-R53	C1/o	XVIII	Q	2	50	2	50	2	50	pF
R54	C1/o	XVIII	Q	10	150	10	150	10	150	pF
R55,R56	C1/o	XVIII	Q	10	175	10	175	10	175	pF
R57	C1/o	XVIII	Q	10	150	10	150	10	150	pF
R58	C1/o	XVIII	Q	10	125	10	125	10	125	pF
R59-R64	C1/o	XVIII	Q	10	150	10	150	10	150	pF
R65	C1/o	XVIII	Q	10	125	10	125	10	125	pF
R66-R68	C1/o	XVIII	Q	12	175	12	175	12	175	pF
R69	C1/o	XVIII	Q	12	150	12	150	12	150	pF
R70	C1/o	XVIII	Q	6	100	6	100	6	100	pF
R71-R74	C1/o	XVIII	Q	8	150	8	150	8	150	pF
R75-R86	C1/o	XVIII	Q	8	125	8	125	8	125	pF
T1	PSTA	XIX	Q	0	600	0	600	0	600	mV
T2	PSTA	XIX	Q	0	1000	0	1000	0	1000	mV

- See footnotes at end of table.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/517-1 for class I, and M28787/517-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

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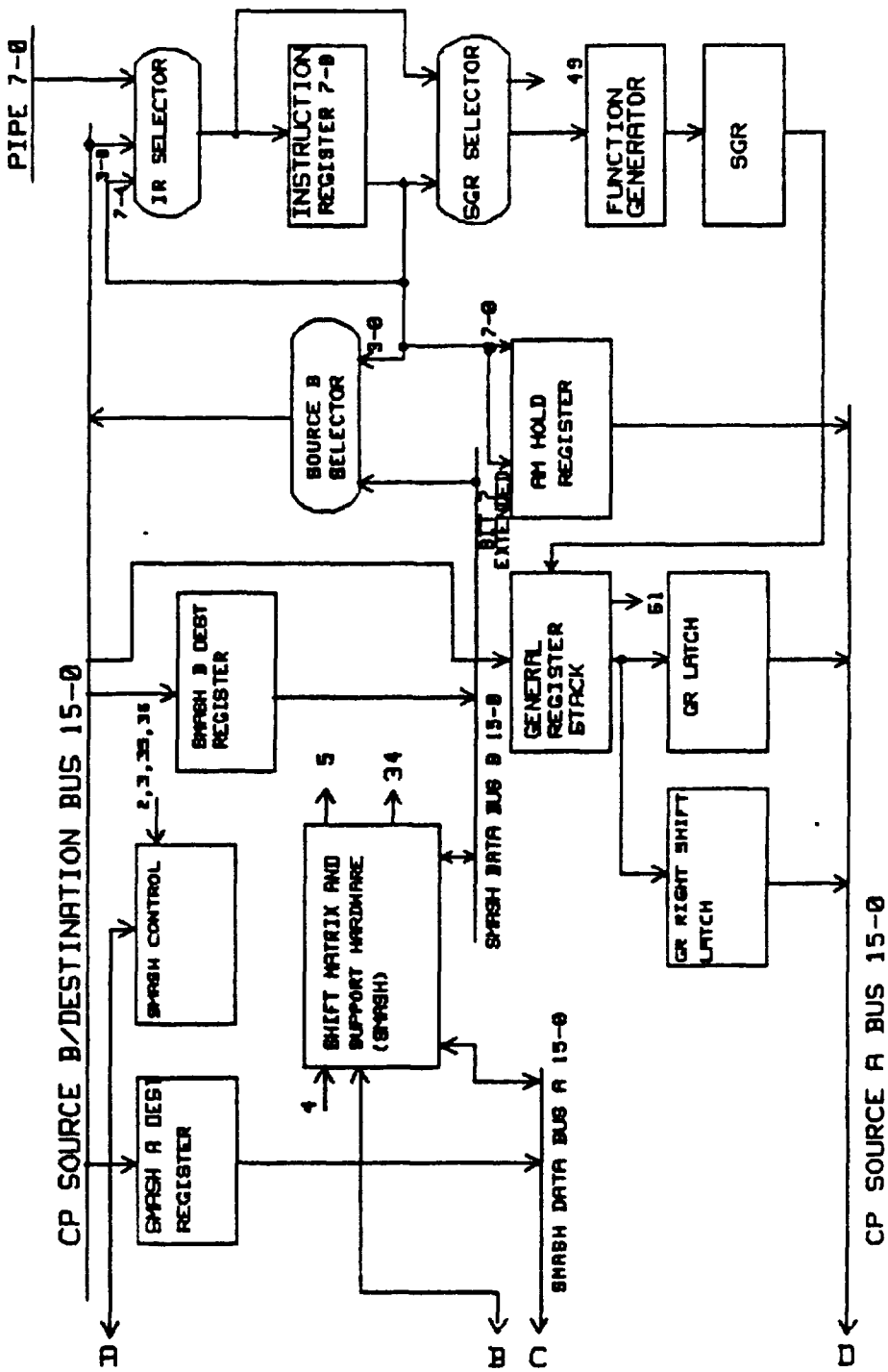


FIGURE 1. CWE functional block diagram.

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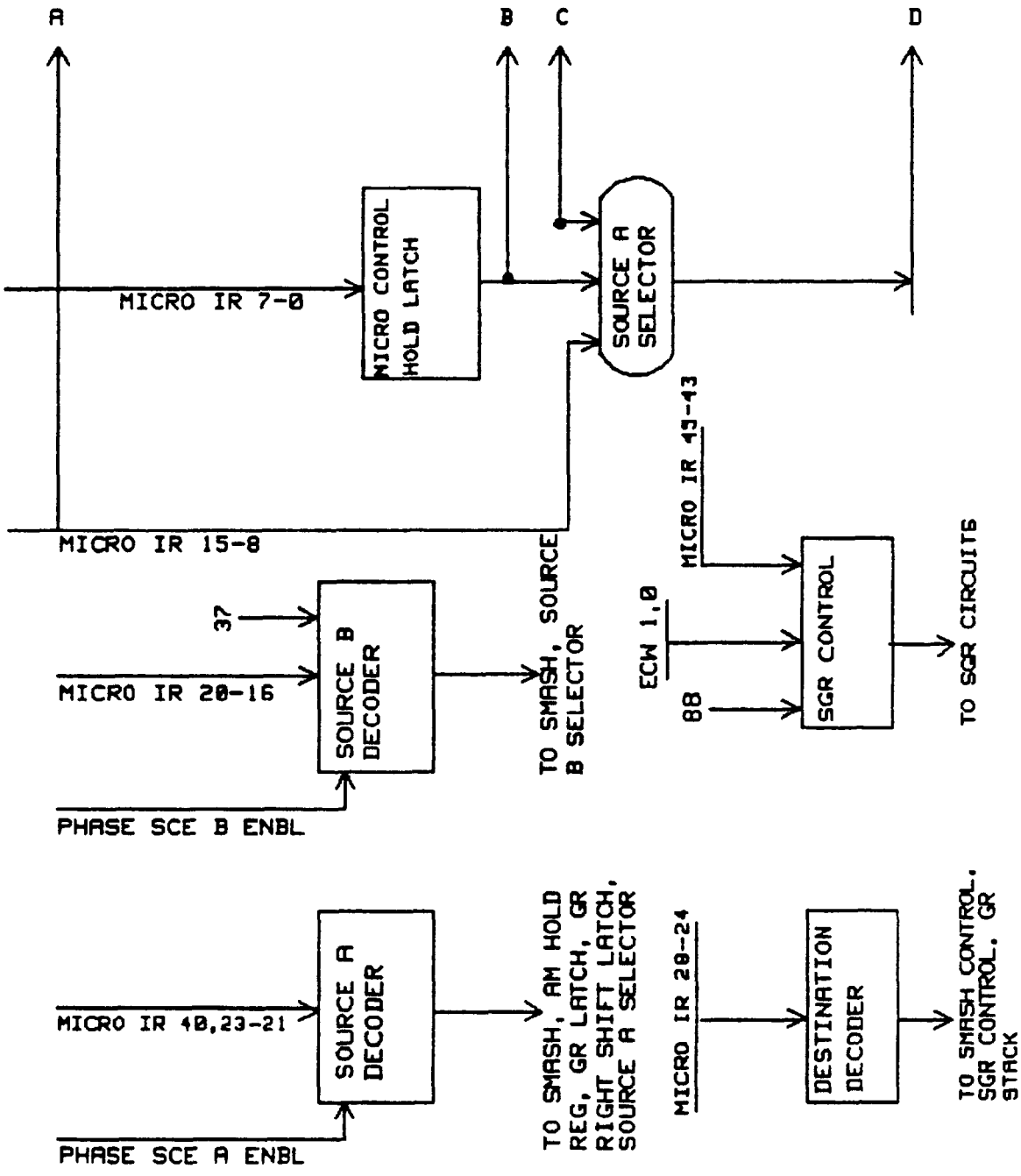


FIGURE 1. CWE functional block diagram - Continued.

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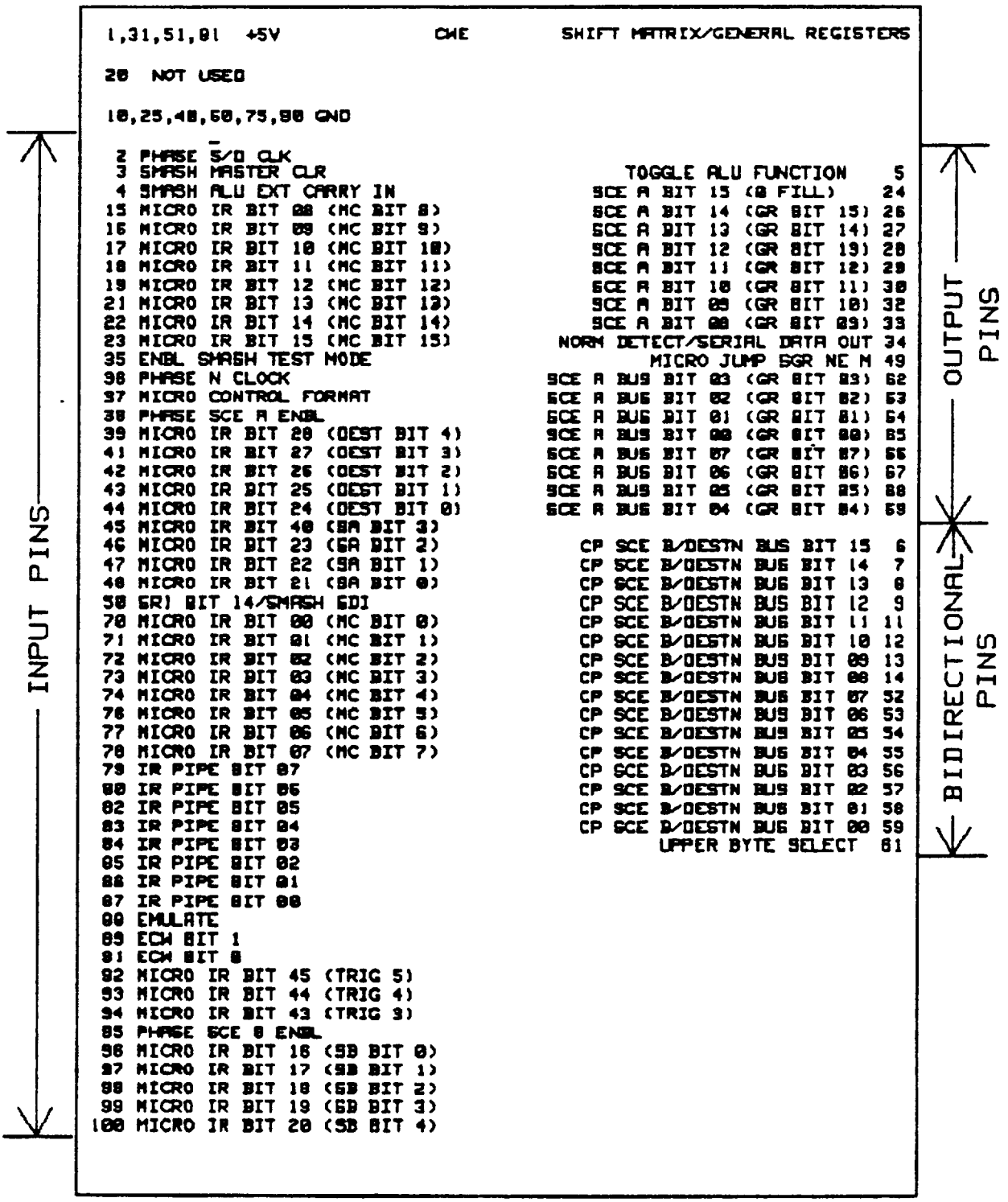


FIGURE 2. CWE pin assignment.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature

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stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

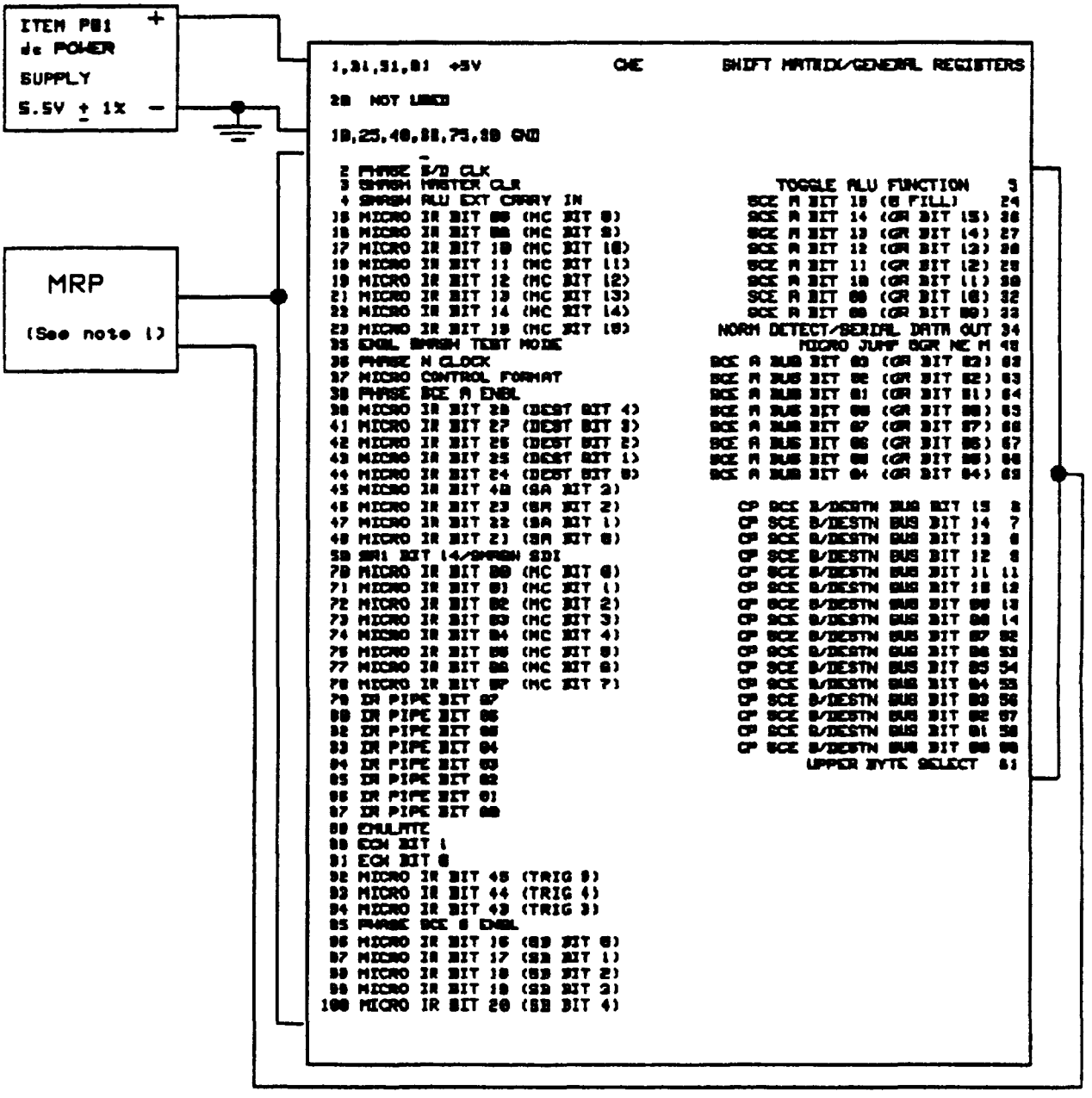
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. CWE operating temperature and component temperature test circuit.

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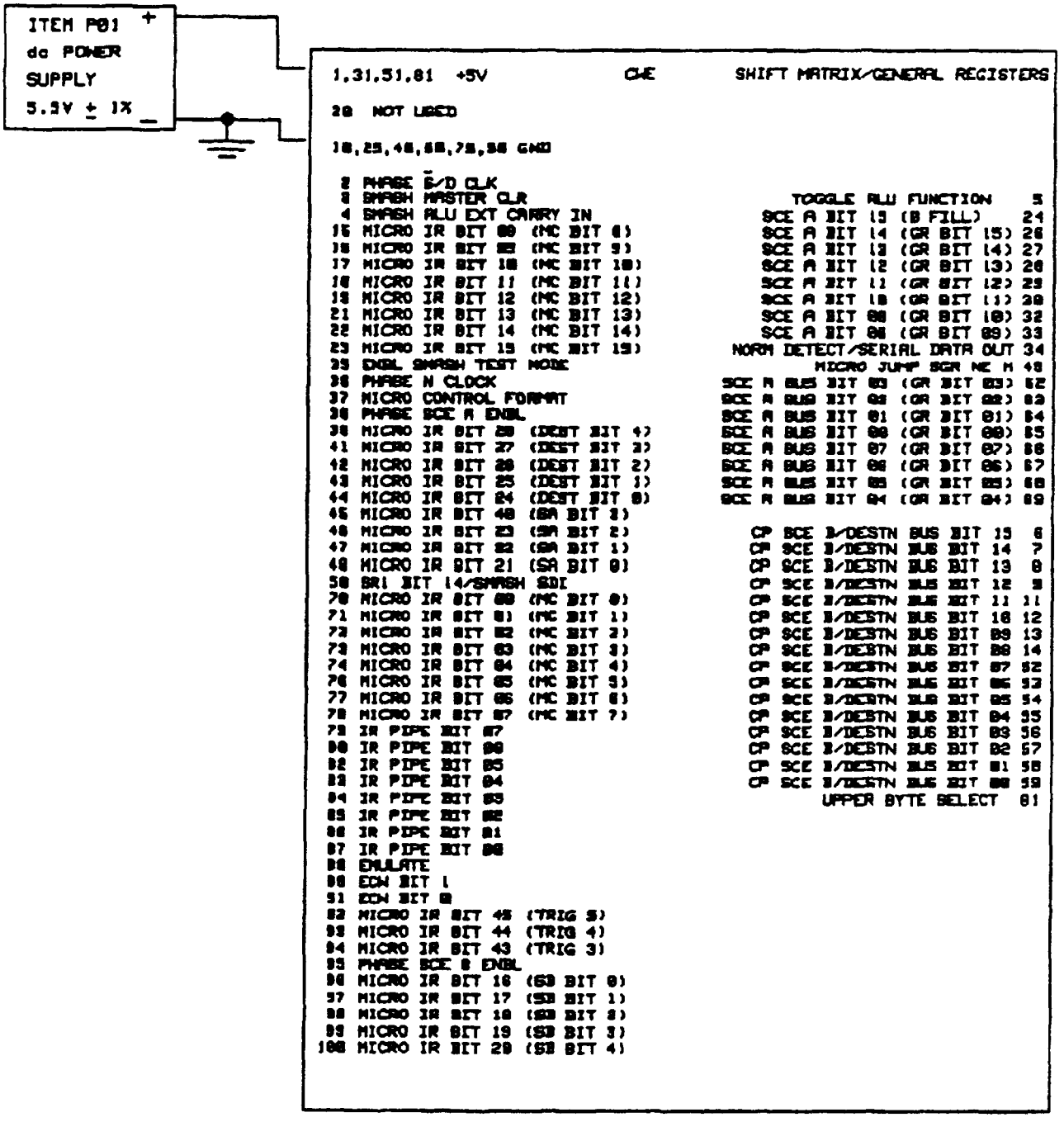
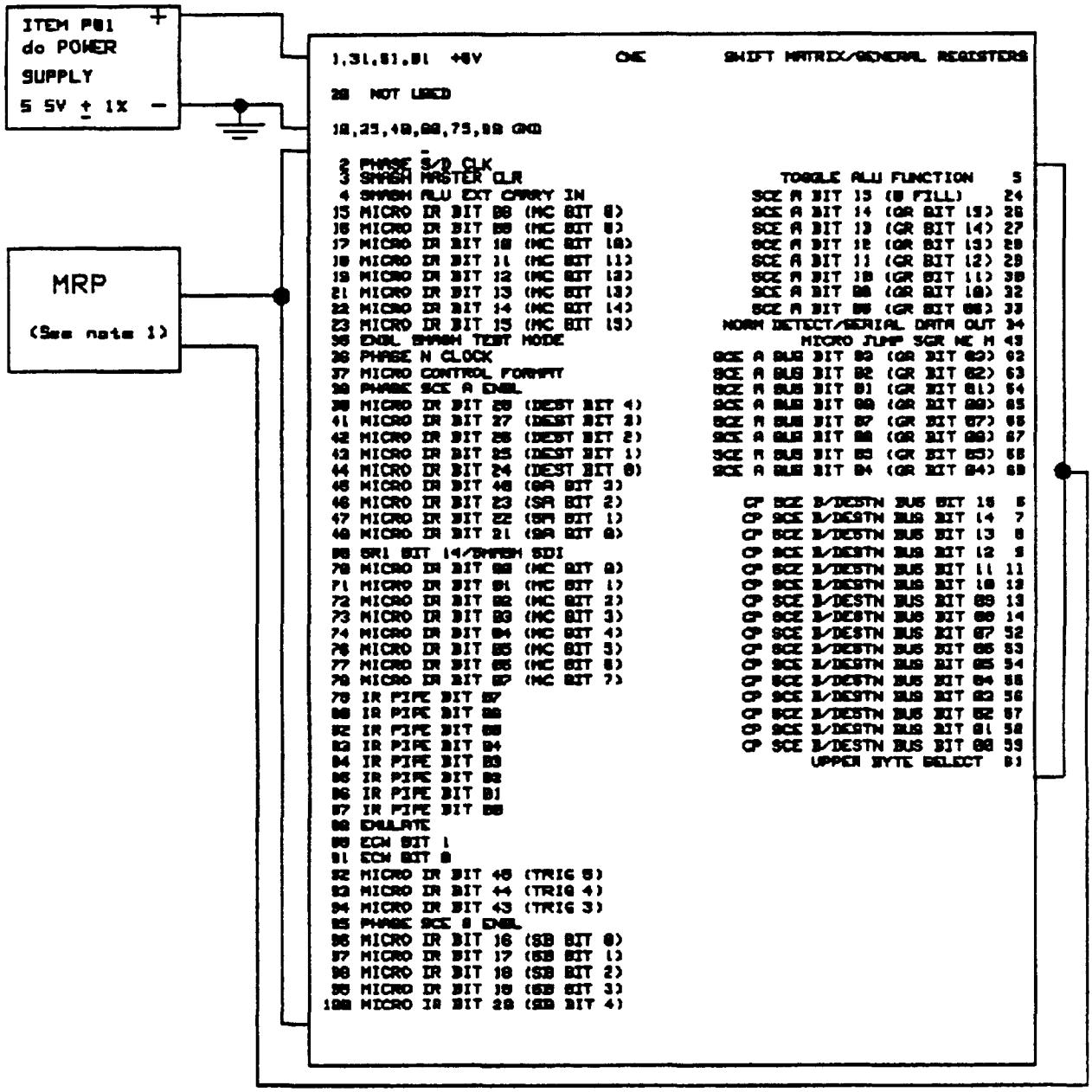


FIGURE 4. CWE vibration test circuit.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. CWE life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 10.56 W (max); 7.08 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
DESTN BIT 15	NA	0.4	2.4	-2.05	0.11	NA	150
DESTN BIT 14	NA	0.4	2.4	-2.05	0.11	NA	175
DESTN BIT 13	NA	0.4	2.4	-2.05	0.11	NA	175
DESTN BIT 12	NA	0.4	2.4	-2.05	0.11	NA	150
DESTN BIT 11	NA	0.4	2.4	-2.05	0.11	NA	125
DESTN BIT 10	NA	0.4	2.4	-2.05	0.11	NA	150
DESTN BIT 09	NA	0.4	2.4	-2.05	0.11	NA	150
DESTN BIT 08	NA	0.4	2.4	-2.05	0.11	NA	150
DESTN BIT 07	NA	0.4	2.4	-2.05	0.11	NA	150
DESTN BIT 06	NA	0.4	2.4	-2.05	0.11	NA	150
DESTN BIT 05	NA	0.4	2.4	-2.05	0.11	NA	150
DESTN BIT 04	NA	0.4	2.4	-2.05	0.11	NA	125
DESTN BIT 03	NA	0.4	2.4	-2.65	0.13	NA	175
DESTN BIT 02	NA	0.4	2.4	-2.65	0.13	NA	175

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
DESTN BIT 01	NA	0.4	2.4	-2.65	0.13	NA	175
DESTN BIT 00	NA	0.4	2.4	-2.65	0.13	NA	150
PHASE CLK	-1.2	0.4	2.4	-6.6	0.17	3.1	125
MSTR CLR	-1.2	0.4	2.4	-0.6	0.02	0.1	75
ALU EXT IN	-1.2	0.4	2.4	-0.6	0.02	0.1	75
MC BIT 8	-1.2	0.4	2.4	-0.85	0.045	1.1	100
MC BIT 9	-1.2	0.4	2.4	-0.85	0.045	1.1	100
MC BIT 10	-1.2	0.4	2.4	-0.85	0.045	1.1	100
MC BIT 11	-1.2	0.4	2.4	-0.85	0.045	1.1	100
MC BIT 12	-1.2	0.4	2.4	-0.85	0.045	1.1	100
MC BIT 13	-1.2	0.4	2.4	-0.85	0.045	1.1	100
MC BIT 14	-1.2	0.4	2.4	-1.2	0.04	0.2	100
MC BIT 15	-1.2	0.4	2.4	-1.2	0.04	0.2	100
EN T MODE	-1.2	0.4	2.4	-0.6	0.02	0.1	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
PH N CLK	-1.2	0.4	2.4	-4.6	0.12	2.1	100
CTRL FRMT	-1.5	0.4	2.4	-0.25	0.025	1.0	75
PH SCE EN	-1.2	0.4	2.4	-0.6	0.02	0.1	50
DEST BIT 4	-1.5	0.4	2.4	-0.4	0.02	0.1	75
DEST BIT 3	-1.5	0.4	2.4	-0.4	0.02	0.1	75
DEST BIT 2	-1.5	0.4	2.4	-0.4	0.02	0.1	75
DEST BIT 1	-1.5	0.4	2.4	-0.4	0.02	0.1	50
DEST BIT 0	-1.5	0.4	2.4	-0.4	0.02	0.1	50
SA BIT 3	-1.2	0.4	2.4	-0.6	0.02	0.1	50
SA BIT 2	-1.2	0.4	2.4	-0.6	0.02	0.1	50
SA BIT 1	-1.2	0.4	2.4	-0.6	0.02	0.1	50
SA BIT 0	-1.2	0.4	2.4	-0.6	0.02	0.1	50
SR1 BIT 14	-1.2	0.4	2.4	-1.2	0.04	0.2	75
MC BIT 0	-1.5	0.4	2.4	-0.2	0.02	0.1	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MC BIT 1	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MC BIT 2	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MC BIT 3	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MC BIT 4	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MC BIT 5	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MC BIT 6	-1.5	0.4	2.4	-0.2	0.02	0.1	50
MC BIT 7	-1.5	0.4	2.4	-0.2	0.02	0.1	50
IR BIT 7	-1.2	0.4	2.4	-0.6	0.02	0.1	50
IR BIT 6	-1.2	0.4	2.4	-0.6	0.02	0.1	50
IR BIT 5	-1.2	0.4	2.4	-0.6	0.02	0.1	50
IR BIT 4	-1.2	0.4	2.4	-0.6	0.02	0.1	50
IR BIT 3	-1.2	0.4	2.4	-0.6	0.02	0.1	50
IR BIT 2	-1.2	0.4	2.4	-0.6	0.02	0.1	50
IR BIT 1	-1.2	0.4	2.4	-0.6	0.02	0.1	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
IR BIT 0	-1.2	0.4	2.4	-0.6	0.02	0.1	50
EMULATE	-1.5	0.4	2.4	-0.25	0.025	1.0	50
ECW BIT 1	-1.5	0.4	2.4	-0.25	0.025	1.0	50
ECW BIT 0	-1.5	0.4	2.4	-0.25	0.025	1.0	50
TRIG 5	-1.5	0.4	2.4	-0.25	0.025	1.0	50
TRIG 4	-1.5	0.4	2.4	-0.25	0.025	1.0	50
TRIG 3	-1.5	0.4	2.4	-0.25	0.025	1.0	50
PH B EN	-1.2	0.4	2.4	-1.2	0.04	0.2	75
SB BIT 0	-1.2	0.4	2.4	-0.6	0.02	0.1	50
SB BIT 1	-1.2	0.4	2.4	-0.6	0.02	0.1	50
SB BIT 2	-1.2	0.4	2.4	-0.6	0.02	0.1	50
SB BIT 3	-1.2	0.4	2.4	-0.6	0.02	0.1	50
SB BIT 4	-1.2	0.4	2.4	-0.6	0.02	0.1	50

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
DESTN BIT 15	2.4	0.5	-0.4	6.0	NA	NA	-60.0	150
DESTN BIT 14	2.4	0.5	-0.4	6.0	NA	NA	-60.0	175
DESTN BIT 13	2.4	0.5	-0.4	6.0	NA	NA	-60.0	175
DESTN BIT 12	2.4	0.5	-0.4	6.0	NA	NA	-60.0	150
DESTN BIT 11	2.4	0.5	-0.4	6.0	NA	NA	-60.0	125
DESTN BIT 10	2.4	0.5	-0.4	6.0	NA	NA	-60.0	150
DESTN BIT 09	2.4	0.5	-0.4	6.0	NA	NA	-60.0	150
DESTN BIT 08	2.4	0.5	-0.4	6.0	NA	NA	-60.0	150
DESTN BIT 07	2.4	0.5	-0.4	6.0	NA	NA	-60.0	150
DESTN BIT 06	2.4	0.5	-0.4	6.0	NA	NA	-60.0	150
DESTN BIT 05	2.4	0.5	-0.4	6.0	NA	NA	-60.0	150
DESTN BIT 04	2.4	0.5	-0.4	6.0	NA	NA	-60.0	125
DESTN BIT 03	2.4	0.5	-0.4	6.0	NA	NA	-60.0	175
DESTN BIT 02	2.4	0.5	-0.4	6.0	NA	NA	-60.0	175

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
DESTN BIT 01	2.4	0.5	-0.4	6.0	NA	NA	-60.0	175
DESTN BIT 00	2.4	0.5	-0.4	6.0	NA	NA	-60.0	150
UP BYTE SEL	2.4	0.5	-0.16	1.6	NA	NA	-20.0	100
ALU FUNCT	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
0 FILL	2.4	0.5	-0.16	1.6	0.17	0.17	-60.0 -30.0	150
GR BIT 15	2.4	0.5	-0.16	1.6	0.17	0.17	-60.0 -30.0	150
GR BIT 14	2.4	0.5	-0.16	1.6	0.17	0.17	-60.0 -30.0	150
GR BIT 13	2.4	0.5	-0.16	1.6	0.17	0.17	-60.0 -30.0	150
GR BIT 12	2.4	0.5	-0.16	1.6	0.17	0.17	-60.0 -30.0	125
GR BIT 11	2.4	0.5	-0.16	1.6	0.17	0.17	-60.0 -30.0	125
GR BIT 10	2.4	0.5	-0.16	1.6	0.17	0.17	-60.0 -30.0	125
GR BIT 09	2.4	0.5	-0.16	1.6	0.17	0.17	-60.0 -30.0	125
SER DATA OUT	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
SGR NE M	2.4	0.5	-0.16	1.6	NA	NA	-15.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
GR BIT 03	2.4	0.5	-0.16	1.6	0.17	0.17	-60.0 -30.0	125
GR BIT 02	2.4	0.5	-0.16	1.6	0.17	0.17	-60.0 -30.0	125
GR BIT 01	2.4	0.5	-0.16	1.6	0.17	0.17	-60.0 -30.0	125
GR BIT 00	2.4	0.5	-0.16	1.6	0.17	0.17	-60.0 -30.0	125
GR BIT 07	2.4	0.5	-0.16	1.6	0.17	0.17	-60.0 -30.0	125
GR BIT 06	2.4	0.5	-0.16	1.6	0.17	0.17	-60.0 -30.0	125
GR BIT 05	2.4	0.5	-0.16	1.6	0.17	0.17	-60.0 -30.0	125
GR BIT 04	2.4	0.5	-0.16	1.6	0.17	0.17	-60.0 -30.0	125

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	PHASE CLK to ALU FUNCT	-	-	-	129	129	129	nS	12
2	tpd	PH B EN to DESTN BIT 00	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 01	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 02	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 03	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 04	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 05	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 06	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 07	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 08	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 09	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 10	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 11	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 12	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 13	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 14	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 15	-	-	-	85	85	85	nS	12
3	tpd	PHASE CLK to DESTN BIT 00	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 01	-	-	-	140	140	140	nS	12

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Check the source to verify that this is the current version before use.

6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tpd	PHASE CLK to DESTN BIT 02	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 03	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 04	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 05	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 06	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 07	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 08	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 09	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 10	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 11	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 12	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 13	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 14	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 15	-	-	-	140	140	140	nS	12
4	tpd	PH SCE EN to GR BIT 00	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 01	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 02	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 03	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 04	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 05	-	-	-	84	84	84	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	PH SCE EN to GR BIT 06	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 07	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to 0 FILL	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 09	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 10	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 11	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 12	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 13	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 14	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 15	-	-	-	84	84	84	nS	12
5	tpd	SR1 BIT 14 to 0 FILL	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 15	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 14	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 13	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 12	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 11	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 10	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 09	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 07	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 06	-	-	-	76	76	76	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tpd	SR1 BIT 14 to GR BIT 05	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 04	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 03	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 02	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 01	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 00	-	-	-	76	76	76	nS	12
6	tpd	PHASE CLK to UP BYTE SEL	-	-	-	96	96	96	nS	12
7	tpd	PHASE CLK to SGR NE M	-	-	-	84	84	84	nS	12
8	tsu	ALU EXT IN to PHASE CLK	60	60	60	-	-	-	nS	12
9	tsu	DESTN BIT 15 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 14 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 13 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 12 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 11 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 10 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 09 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 08 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 07 to PHASE CLK	52	52	52	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
9	tsu	DESTN BIT 06 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 05 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 04 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 03 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 02 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 01 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 00 to PHASE CLK	52	52	52	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 15	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 14	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 13	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 12	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 11	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 10	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 09	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 08	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 07	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 06	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 05	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 04	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 03	17	17	17	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
10	th	PH N CLK to DESTN BIT 02	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 01	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 00	17	17	17	-	-	-	nS	12
11	tsu	EMULATE to PHASE CLK	87	87	87	-	-	-	nS	12
11	tsu	ECW BIT 1 to PHASE CLK	87	87	87	-	-	-	nS	12
11	tsu	ECW BIT 0 to PHASE CLK	87	87	87	-	-	-	nS	12
12	tsu	IR BIT 7 to PHASE CLK	46	46	46	-	-	-	nS	12
12	tsu	IR BIT 6 to PHASE CLK	46	46	46	-	-	-	nS	12
12	tsu	IR BIT 5 to PHASE CLK	46	46	46	-	-	-	nS	12
12	tsu	IR BIT 4 to PHASE CLK	46	46	46	-	-	-	nS	12
12	tsu	IR BIT 3 to PHASE CLK	46	46	46	-	-	-	nS	12
12	tsu	IR BIT 2 to PHASE CLK	46	46	46	-	-	-	nS	12
12	tsu	IR BIT 1 to PHASE CLK	46	46	46	-	-	-	nS	12
12	tsu	IR BIT 0 to PHASE CLK	46	46	46	-	-	-	nS	12

6.4 Module function. This module (CWE) contains the shift matrix and two sets of 16 16-bit general registers. The shift matrix performs high speed shifts, normalizations, reverse register, iterative multiply and divide, and square root operations. The shift matrix uses both the source A bus and the source B/destination bus for iterative operations. The general registers are general purpose registers used by the macroinstructions. A general register can be used as 1) an accumulater for arithmetic, shift, and logical functions; 2) an

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index register for address and operand modification; or 3) a temporary storage location for addresses, operands, etc. The macroinstructions operation and format codes define the use of the register and the a and m designators select the register or registers in a set. The a or m designator is loaded into the 4-bit general register address register to select a general register within a set. A set is selected by status register 1, bit 14.

6.4.1 Shift Matrix And Support Hardware (SMASH). SMASH is an integrated circuit that provides:

(a) A 16-bit shift matrix.

(b) Multiply algorithm with 16-bit operands.

(c) Divide algorithm with a 32-bit dividend and 16-bit divisor.

(d) Square root algorithm with a 30-bit operand.

(e) Normalize detector for 16- or 32-bit numbers.

(f) Output bit masking.

Operands are transferred to/from SMASH via the SMASH data bus A and data bus B. SMASH is controlled by a format 3 micro-instruction with MC = F. Microinstruction bits 9-0 (MCU) determine the operation.

6.4.2 SMASH A and B destination registers. The destination registers hold data for the SMASH A and B ports. The A destination register is enabled by a microinstruction with SA = 8. The B destination register is enabled by a microinstruction with D = 08.

6.4.3 SMASH control. The SMASH control Programmable Array Logic (PAL) decodes microinstruction bits 15-8 to provide control signals to SMASH.

6.4.4 Source A selector. The 16-bit selector transfers either the format 2 microinstruction K-field or SMASH data bus A to the Central Processor (CP) source A bus. The K-field is selected by a microinstruction with SA = F. The selector is enabled by a microinstruction with SA = 8 or F.

6.4.5 Source B selector. The 16-bit selector transfers

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either the SMASH data bus B or macroinstruction m-field (with zero fill in the upper bits) to the source B/destination bus. The m-field is selected by a microinstruction with SB = 1D. The selector is enabled by a microinstruction with SB = 1C or 1D.

6.4.6 General Register (GR) stack. The general register stack contains two sets (set 0 and set 1) of 16 16-bit general registers. A general register within a set is addressed by the 4-bit general register address register. A set is selected by status register 1 bit 14 (bit 14 = 0, select set 0: bit 14 = 1, select set 1). The addressed general register is loaded from the source B/destination bus by a microinstruction with D = 0B. The general register output is placed on the source A bus via the GR latch or GR right shift latch. The GR latch is enabled by a microinstruction with SA = C or D. The GR right shift latch shifts the general register output right one bit position and places a zero in bit 15. The latch is enabled by a microinstruction with SA = E.

6.4.7 Select General Register (SGR) circuits. The SGR circuits consist of SGR control, function generator, and SGR. SGR holds the 4-bit address of a general register within a set. SGR control interprets the microinstruction SGR field (bits 45-43) and the Emulation Control Word (ECW) SGR control bits to control the SGR selector, function generator, and SGR. The 4-bit SGR selector selects one of four inputs; the instruction register a-field, m-field, IR selector bit 7-4, or IR selector bits 3-0. The output passes through the function generator to SGR. The function generator adds 0, 1, 2 or 3 to the 4-bit address depending on the interpretation of the microinstruction SGR field or ECW SGR control bits.

6.4.8 Instruction Register (IR) selector. The 8-bit IR selector selects either pipe bits 7-0 or IR bits 7-4 and source B/destination bits 3-0. IR bits 7-4 and source B/destination bits 3-0 are selected by a microinstruction with SB = 0A, otherwise pipe bits 7-0 are selected.

6.4.9 Instruction register 7-0. The 8-bit instruction register holds the output from the IR selector. The register is loaded by a microinstruction with the SGR field = 0 and SB = 0A or by an emulate microinstruction with bit 43 = 0.

6.4.10 Macroinstruction a- and m-fields (AM) hold register. The 8-bit register holds the macroinstruction a- and m-fields. The register output is enabled by a microinstruction with SA = 9.

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6.4.11 Description of I/O pins.

Signal name	Pin number	I/O	Description
DESTN BIT 15	6	I/O	CP SCE B/DESTN BUS BIT 15
DESTN BIT 14	7	I/O	CP SCE B/DESTN BUS BIT 14
DESTN BIT 13	8	I/O	CP SCE B/DESTN BUS BIT 13
DESTN BIT 12	9	I/O	CP SCE B/DESTN BUS BIT 12
DESTN BIT 11	11	I/O	CP SCE B/DESTN BUS BIT 11
DESTN BIT 10	12	I/O	CP SCE B/DESTN BUS BIT 10
DESTN BIT 09	13	I/O	CP SCE B/DESTN BUS BIT 09
DESTN BIT 08	14	I/O	CP SCE B/DESTN BUS BIT 08
DESTN BIT 07	52	I/O	CP SCE B/DESTN BUS BIT 07
DESTN BIT 06	53	I/O	CP SCE B/DESTN BUS BIT 06
DESTN BIT 05	54	I/O	CP SCE B/DESTN BUS BIT 05
DESTN BIT 04	55	I/O	CP SCE B/DESTN BUS BIT 04
DESTN BIT 03	56	I/O	CP SCE B/DESTN BUS BIT 03
DESTN BIT 02	57	I/O	CP SCE B/DESTN BUS BIT 02
DESTN BIT 01	58	I/O	CP SCE B/DESTN BUS BIT 01
DESTN BIT 00	59	I/O	CP SCE B/DESTN BUS BIT 00
UP BYTE SEL	61	I/O	UPPER BYTE SELECT
PHASE CLK	2	I	PHASE $\overline{S}$/D CLK
MSTR CLR	3	I	SMASH MASTER CLR
ALU EXT IN	4	I	SMASH ALU EXT CARRY IN

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6.4.11 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MC BIT 8	15	I	MICRO IR BIT 08 (MC BIT 8)
MC BIT 9	16	I	MICRO IR BIT 09 (MC BIT 9)
MC BIT 10	17	I	MICRO IR BIT 10 (MC BIT 10)
MC BIT 11	18	I	MICRO IR BIT 11 (MC BIT 11)
MC BIT 12	19	I	MICRO IR BIT 12 (MC BIT 12)
MC BIT 13	21	I	MICRO IR BIT 13 (MC BIT 13)
MC BIT 14	22	I	MICRO IR BIT 14 (MC BIT 14)
MC BIT 15	23	I	MICRO IR BIT 15 (MC BIT 15)
EN T MODE	35	I	ENABLE SMASH TEST MODE
PH N CLK	36	I	PHASE N CLOCK
CTRL FRMT	37	I	MICRO CONTROL FORMAT
PH SCE EN	38	I	PHASE SCE A ENABLE
DEST BIT 4	39	I	MICRO IR BIT 28 (DEST BIT 4)
DEST BIT 3	41	I	MICRO IR BIT 27 (DEST BIT 3)
DEST BIT 2	42	I	MICRO IR BIT 26 (DEST BIT 2)
DEST BIT 1	43	I	MICRO IR BIT 25 (DEST BIT 1)
DEST BIT 0	44	I	MICRO IR BIT 24 (DEST BIT 0)
SA BIT 3	45	I	MICRO IR BIT 40 (SA BIT 3)
SA BIT 2	46	I	MICRO IR BIT 23 (SA BIT 2)
SA BIT 1	47	I	MICRO IR BIT 22 (SA BIT 1)

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6.4.11 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
SA BIT 0	48	I	MICRO IR BIT 21 (SA BIT 0)
SR1 BIT 14	50	I	SR1 BIT 14/SMASH SDI
MC BIT 0	70	I	MICRO IR BIT 00 (MC BIT 0)
MC BIT 1	71	I	MICRO IR BIT 01 (MC BIT 1)
MC BIT 2	72	I	MICRO IR BIT 02 (MC BIT 2)
MC BIT 3	73	I	MICRO IR BIT 03 (MC BIT 3)
MC BIT 4	74	I	MICRO IR BIT 04 (MC BIT 4)
MC BIT 5	76	I	MICRO IR BIT 05 (MC BIT 5)
MC BIT 6	77	I	MICRO IR BIT 06 (MC BIT 6)
MC BIT 7	78	I	MICRO IR BIT 07 (MC BIT 7)
IR BIT 7	79	I	IR PIPE BIT 07
IR BIT 6	80	I	IR PIPE BIT 06
IR BIT 5	82	I	IR PIPE BIT 05
IR BIT 4	83	I	IR PIPE BIT 04
IR BIT 3	84	I	IR PIPE BIT 03
IR BIT 2	85	I	IR PIPE BIT 02
IR BIT 1	86	I	IR PIPE BIT 01
IR BIT 0	87	I	IR PIPE BIT 00
EMULATE	88	I	EMULATE
ECW BIT 1	89	I	ECW BIT 1

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6.4.11 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
ECW BIT 0	91	I	ECW BIT 0
TRIG 5	92	I	MICRO IR BIT 45 (TRIG 5)
TRIG 4	93	I	MICRO IR BIT 44 (TRIG 4)
TRIG 3	94	I	MICRO IR BIT 43 (TRIG 3)
PH B EN	95	I	PHASE SCE B ENABLE
SB BIT 0	96	I	MICRO IR BIT 16 (SB BIT 0)
SB BIT 1	97	I	MICRO IR BIT 17 (SB BIT 1)
SB BIT 2	98	I	MICRO IR BIT 18 (SB BIT 2)
SB BIT 3	99	I	MICRO IR BIT 19 (SB BIT 3)
SB BIT 4	100	I	MICRO IR BIT 20 (SB BIT 4)
ALU FUNCT	5	O	TOGGLE ALU FUNCTION
0 FILL	24	O	SCE A BIT 15 (0 FILL)
GR BIT 15	26	O	SCE A BIT 14 (GR BIT 15)
GR BIT 14	27	O	SCE A BIT 13 (GR BIT 14)
GR BIT 13	28	O	SCE A BIT 12 (GR BIT 13)
GR BIT 12	29	O	SCE A BIT 11 (GR BIT 12)
GR BIT 11	30	O	SCE A BIT 10 (GR BIT 11)
GR BIT 10	32	O	SCE A BIT 09 (GR BIT 10)
GR BIT 09	33	O	SCE A BIT 08 (GR BIT 09)
SER DATA OUT	34	O	NORM DETECT/SERIAL DATA OUT

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6.4.11 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
SGR NE M	49	O	MICRO JUMP SGR NE M
GR BIT 03	62	O	SCE A BUS BIT 03 (GR BIT 03)
GR BIT 02	63	O	SCE A BUS BIT 02 (GR BIT 02)
GR BIT 01	64	O	SCE A BUS BIT 01 (GR BIT 01)
GR BIT 00	65	O	SCE A BUS BIT 00 (GR BIT 00)
GR BIT 07	66	O	SCE A BUS BIT 07 (GR BIT 07)
GR BIT 06	67	O	SCE A BUS BIT 06 (GR BIT 06)
GR BIT 05	68	O	SCE A BUS BIT 05 (GR BIT 05)
GR BIT 04	69	O	SCE A BUS BIT 04 (GR BIT 04)

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N710)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect A to	Connect B to	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	C,20	Frame	10	INF	MOhm	1-8
A2	Isolation, unused to used pins	C	20	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1,A2
Isolation measurement	NA	± 500 KOhm	A1,A2

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

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NOTES: - Continued.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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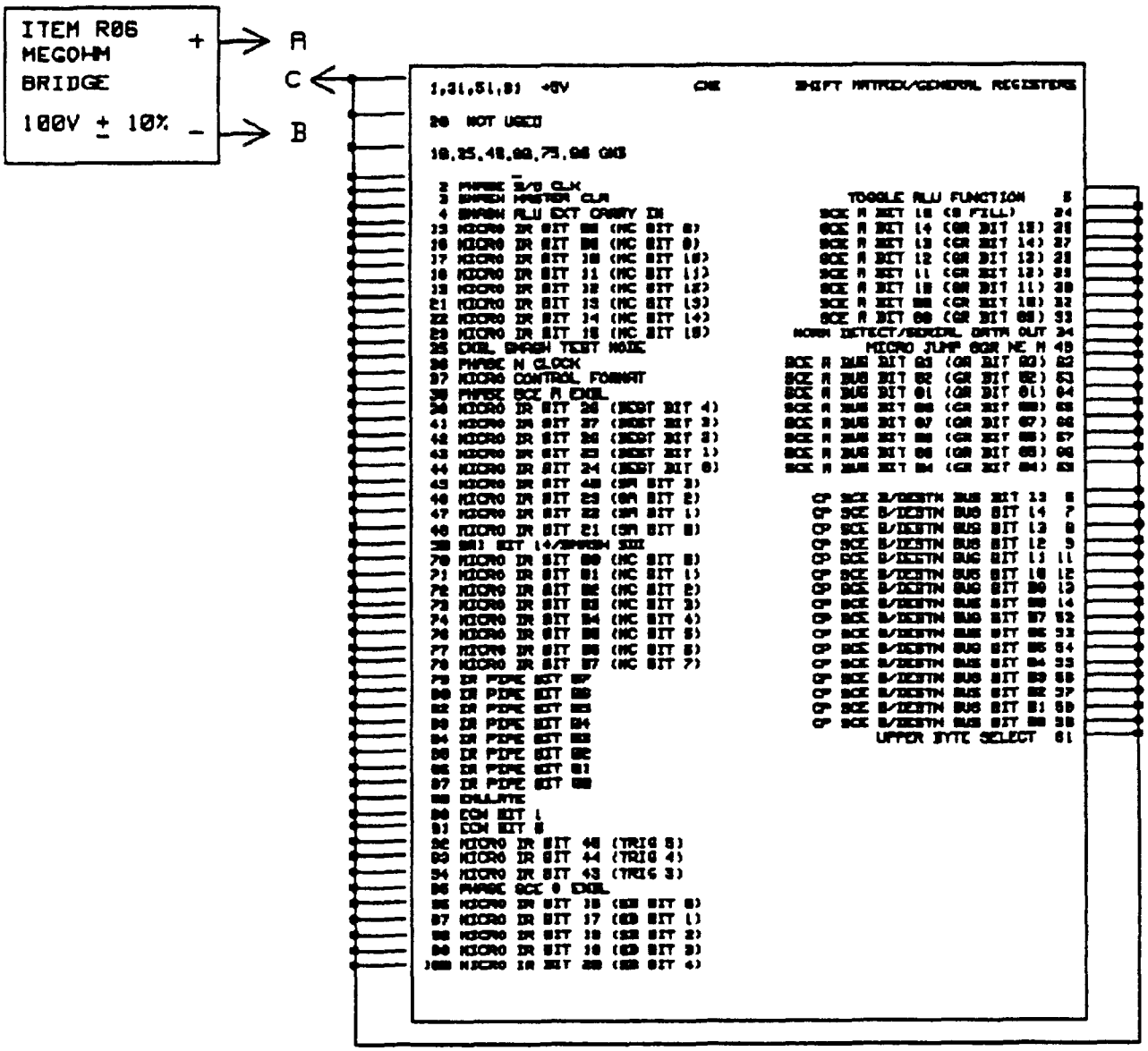


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to	Connect B to	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, +5V	1	31				
B6	Continuity, +5V	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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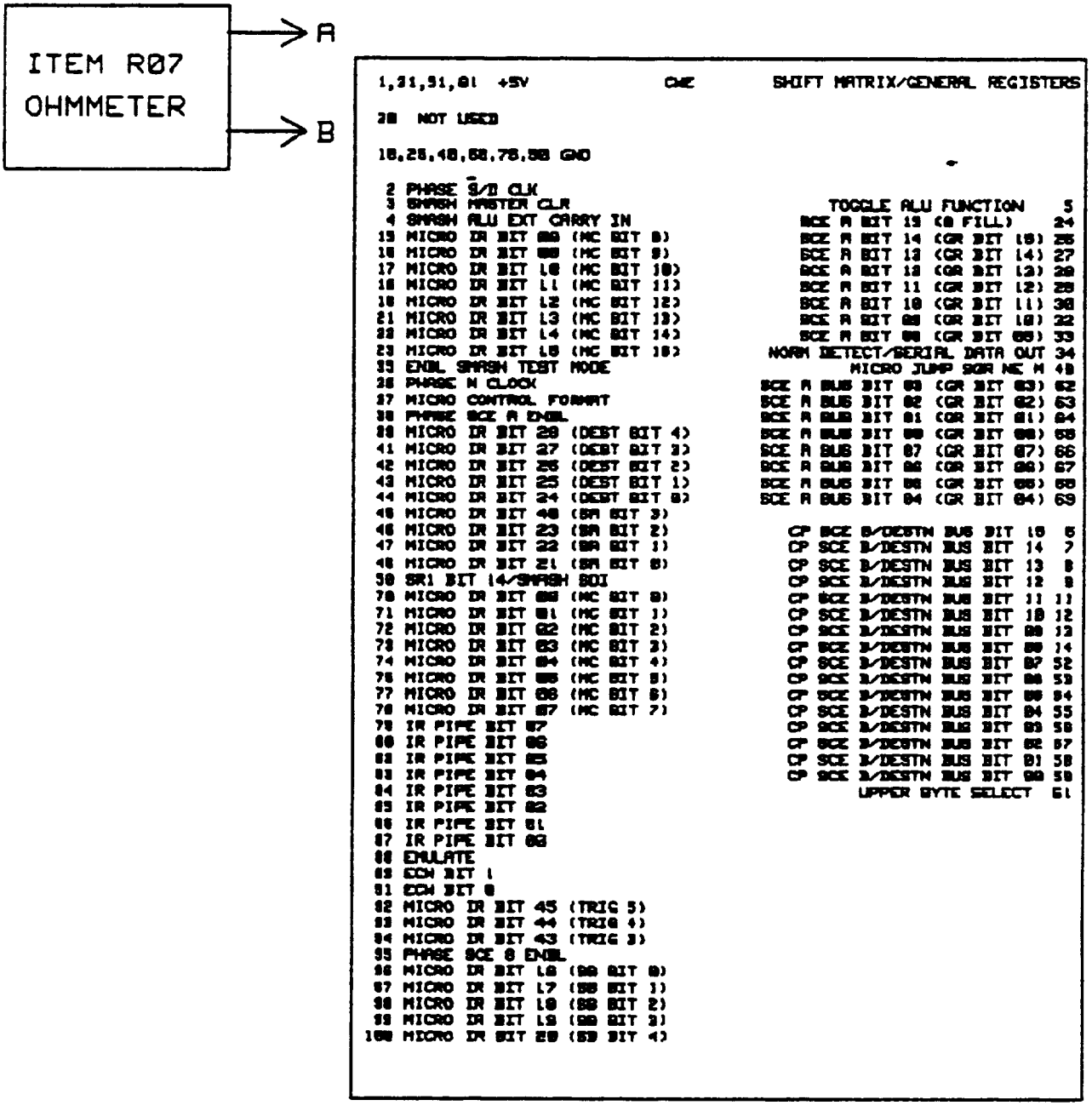


FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.00 V	+0 mV,-70 mV	C
Power supply current limit	3.46 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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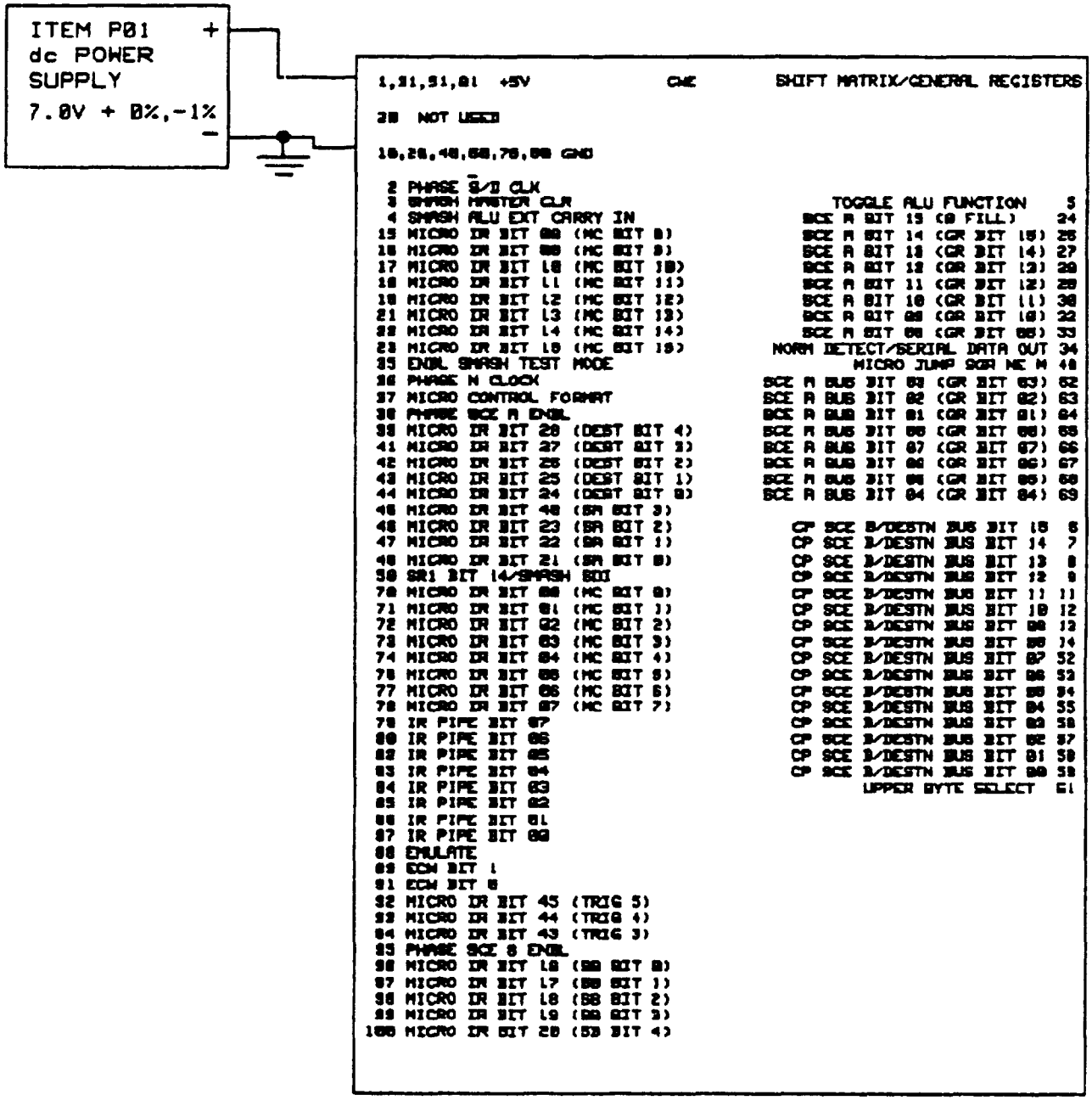


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	PHASE CLK	2	5.5	0.0	3.1	mA	1-3
D2	MSTR CLR	3	5.5	0.0	0.1	mA	1-3
D3	ALU EXT IN	4	5.5	0.0	0.1	mA	1-3
D4	MC BIT 8	15	5.5	0.0	1.1	mA	1-3
D5	MC BIT 9	16					
D6	MC BIT 10	17					
D7	MC BIT 11	18					
D8	MC BIT 12	19					
D9	MC BIT 13	21	5.5	0.0	1.1	mA	1-3
D10	MC BIT 14	22	5.5	0.0	0.2	mA	1-3
D11	MC BIT 15	23	5.5	0.0	0.2	mA	1-3
D12	EN T MODE	35	5.5	0.0	0.1	mA	1-3
D13	PH N CLK	36	5.5	0.0	2.1	mA	1-3
D14	CTRL FRMT	37	5.5	0.0	1.0	mA	1-3
D15	PH SCE EN	38	5.5	0.0	0.1	mA	1-3
D16	DEST BIT 4	39					
D17	DEST BIT 3	41					
D18	DEST BIT 2	42					
D19	DEST BIT 1	43					
D20	DEST BIT 0	44					
D21	SA BIT 3	45	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
D22	SA BIT 2	46	5.5	0.0	0.1	mA	1-3
D23	SA BIT 1	47					
D24	SA BIT 0	48	5.5	0.0	0.1	mA	1-3
D25	SR1 BIT 14	50	5.5	0.0	0.2	mA	1-3
D26	MC BIT 0	70	5.5	0.0	0.1	mA	1-3
D27	MC BIT 1	71					
D28	MC BIT 2	72					
D29	MC BIT 3	73					
D30	MC BIT 4	74					
D31	MC BIT 5	76					
D32	MC BIT 6	77					
D33	MC BIT 7	78					
D34	IR BIT 7	79					
D35	IR BIT 6	80					
D36	IR BIT 5	82					
D37	IR BIT 4	83					
D38	IR BIT 3	84					
D39	IR BIT 2	85					
D40	IR BIT 1	86					
D41	IR BIT 0	87	5.5	0.0	0.1	mA	1-3
D42	EMULATE	88	5.5	0.0	1.0	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D43	ECW BIT 1	89	5.5	0.0	1.0	mA	1-3
D44	ECW BIT 0	91					
D45	TRIG 5	92					
D46	TRIG 4	93					
D47	TRIG 3	94	5.5	0.0	1.0	mA	1-3
D48	PH B EN	95	5.5	0.0	0.2	mA	1-3
D49	SB BIT 0	96	5.5	0.0	0.1	mA	1-3
D50	SB BIT 1	97					
D51	SB BIT 2	98					
D52	SB BIT 3	99					
D53	SB BIT 4	100	5.5	0.0	0.1	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D53
Power supply current limit	2.9 A	NA	D1-D53
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D53
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	D1-D53
low, (VL)	0.0 V	± 100 mV	D1-D53
Input current measurement	NA	± 155 uA	D1
	NA	± 5 uA	D2,D3
	NA	± 55 uA	D4-D9
	NA	± 10 uA	D10,D11
	NA	± 5 uA	D12
	NA	± 105 uA	D13
	NA	± 50 uA	D14
	NA	± 5 uA	D15-D24
	NA	± 10 uA	D25
	NA	± 5 uA	D26-D41
	NA	± 50 uA	D42-D47
	NA	± 10 uA	D48
	NA	± 5 uA	D49-D53

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	ALU FUNCT	5	0.0	-150.0	-60.0	mA	1-4
E2	DESTN BIT 15	6					
E3	DESTN BIT 14	7					
E4	DESTN BIT 13	8					
E5	DESTN BIT 12	9					
E6	DESTN BIT 11	11					
E7	DESTN BIT 10	12					
E8	DESTN BIT 09	13					
E9	DESTN BIT 08	14					
E10	0 FILL	24					
E11	0 FILL	24	0.0	-150.0	-60.0	mA	1-4
E12	0 FILL	24	0.0	-220.0	-30.0	mA	1-4
E13	GR BIT 15	26	0.0	-150.0	-60.0	mA	1-4
E14	GR BIT 15	26					
E15	GR BIT 15	26	0.0	-150.0	-60.0	mA	1-4
E16	GR BIT 15	26	0.0	-220.0	-30.0	mA	1-4
E17	GR BIT 14	27	0.0	-150.0	-60.0	mA	1-4
E18	GR BIT 14	27					
E19	GR BIT 14	27	0.0	-150.0	-60.0	mA	1-4
E20	GR BIT 14	27	0.0	-220.0	-30.0	mA	1-4
E21	GR BIT 13	28	0.0	-150.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E22	GR BIT 13	28	0.0	-150.0	-60.0	mA	1-4
E23	GR BIT 13	28	0.0	-150.0	-60.0	mA	1-4
E24	GR BIT 13	28	0.0	-220.0	-30.0	mA	1-4
E25	GR BIT 12	29	0.0	-150.0	-60.0	mA	1-4
E26	GR BIT 12	29	″	″	″	″	″
E27	GR BIT 12	29	0.0	-150.0	-60.0	mA	1-4
E28	GR BIT 12	29	0.0	-220.0	-30.0	mA	1-4
E29	GR BIT 11	30	0.0	-150.0	-60.0	mA	1-4
E30	GR BIT 11	30	″	″	″	″	″
E31	GR BIT 11	30	0.0	-150.0	-60.0	mA	1-4
E32	GR BIT 11	30	0.0	-220.0	-30.0	mA	1-4
E33	GR BIT 10	32	0.0	-150.0	-60.0	mA	1-4
E34	GR BIT 10	32	″	″	″	″	″
E35	GR BIT 10	32	0.0	-150.0	-60.0	mA	1-4
E36	GR BIT 10	32	0.0	-220.0	-30.0	mA	1-4
E37	GR BIT 09	33	0.0	-150.0	-60.0	mA	1-4
E38	GR BIT 09	33	″	″	″	″	″
E39	GR BIT 09	33	0.0	-150.0	-60.0	mA	1-4
E40	GR BIT 09	33	0.0	-220.0	-30.0	mA	1-4
E41	SER DATA OUT	34	0.0	-150.0	-60.0	mA	1-4
E42	SGR NE M	49	0.0	-100.0	-15.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E43	DESTN BIT 07	52	0.0	-150.0	-60.0	mA	1-4
E44	DESTN BIT 06	53					
E45	DESTN BIT 05	54					
E46	DESTN BIT 04	55					
E47	DESTN BIT 03	56					
E48	DESTN BIT 02	57					
E49	DESTN BIT 01	58					
E50	DESTN BIT 00	59	0.0	-150.0	-60.0	mA	1-4
E51	UP BYTE SEL	61	0.0	-90.0	-20.0	mA	1-4
E52	UP BYTE SEL	61	0.0	-90.0	-20.0	mA	1-4
E53	GR BIT 03	62	0.0	-150.0	-60.0	mA	1-4
E54	GR BIT 03	62					
E55	GR BIT 03	62	0.0	-150.0	-60.0	mA	1-4
E56	GR BIT 03	62	0.0	-220.0	-30.0	mA	1-4
E57	GR BIT 02	63	0.0	-150.0	-60.0	mA	1-4
E58	GR BIT 02	63					
E59	GR BIT 02	63	0.0	-150.0	-60.0	mA	1-4
E60	GR BIT 02	63	0.0	-220.0	-30.0	mA	1-4
E61	GR BIT 01	64	0.0	-150.0	-60.0	mA	1-4
E62	GR BIT 01	64					
E63	GR BIT 01	64	0.0	-150.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E64	GR BIT 01	64	0.0	-220.0	-30.0	mA	1-4
E65	GR BIT 00	65	0.0	-150.0	-60.0	mA	1-4
E66	GR BIT 00	65					
E67	GR BIT 00	65	0.0	-150.0	-60.0	mA	1-4
E68	GR BIT 00	65	0.0	-220.0	-30.0	mA	1-4
E69	GR BIT 07	66	0.0	-150.0	-60.0	mA	1-4
E70	GR BIT 07	66					
E71	GR BIT 07	66	0.0	-150.0	-60.0	mA	1-4
E72	GR BIT 07	66	0.0	-220.0	-30.0	mA	1-4
E73	GR BIT 06	67	0.0	-150.0	-60.0	mA	1-4
E74	GR BIT 06	67					
E75	GR BIT 06	67	0.0	-150.0	-60.0	mA	1-4
E76	GR BIT 06	67	0.0	-220.0	-30.0	mA	1-4
E77	GR BIT 05	68	0.0	-150.0	-60.0	mA	1-4
E78	GR BIT 05	68					
E79	GR BIT 05	68	0.0	-150.0	-60.0	mA	1-4
E80	GR BIT 05	68	0.0	-220.0	-30.0	mA	1-4
E81	GR BIT 04	69	0.0	-150.0	-60.0	mA	1-4
E82	GR BIT 04	69					
E83	GR BIT 04	69	0.0	-150.0	-60.0	mA	1-4
E84	GR BIT 04	69	0.0	-220.0	-30.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E84
Power supply current limit	2.9 A	NA	E1-E84
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E84
Pattern voltage high, (VH)	3.0 V	± 100 mV	E1-E84
low, (VL)	0.0 V	± 100 mV	E1-E84
Output current measurement	NA	± 4.50 mA	E1-E11
	NA	± 9.50 mA	E12
	NA	± 4.50 mA	E13-E15
	NA	± 9.50 mA	E16
	NA	± 4.50 mA	E17-E19
	NA	± 9.50 mA	E20
	NA	± 4.50 mA	E21-E23
	NA	± 9.50 mA	E24
	NA	± 4.50 mA	E25-E27
	NA	± 9.50 mA	E28
	NA	± 4.50 mA	E29-E31
	NA	± 9.50 mA	E32
	NA	± 4.50 mA	E33-E35
	NA	± 9.50 mA	E36
	NA	± 4.50 mA	E37-E39
	NA	± 9.50 mA	E40
	NA	± 4.50 mA	E41
	NA	± 4.25 mA	E42
	NA	± 4.50 mA	E43-E50
	NA	± 3.50 mA	E51,E52
	NA	± 4.50 mA	E53-E55
	NA	± 9.50 mA	E56

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output current measurement	NA	± 4.50 mA	E57-E59
	NA	± 9.50 mA	E60
	NA	± 4.50 mA	E61-E63
	NA	± 9.50 mA	E64
	NA	± 4.50 mA	E65-E67
	NA	± 9.50 mA	E68
	NA	± 4.50 mA	E69-E71
	NA	± 9.50 mA	E72
	NA	± 4.50 mA	E73-E75
	NA	± 9.50 mA	E76
	NA	± 4.50 mA	E77-E79
	NA	± 9.50 mA	E80
	NA	± 4.50 mA	E81-E83
	NA	± 9.50 mA	E84

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

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NOTES: - Continued.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	PHASE CLK	2	2.4	-6.6	0.17	mA	1-3
F2	MSTR CLR	3	2.4	-0.6	0.02	mA	1-3
F3	ALU EXT IN	4	2.4	-0.6	0.02	mA	1-3
F4	DESTN BIT 15	6	2.4	-2.05	0.11	mA	1-3
F5	DESTN BIT 14	7					
F6	DESTN BIT 13	8					
F7	DESTN BIT 12	9					
F8	DESTN BIT 11	11					
F9	DESTN BIT 10	12					
F10	DESTN BIT 09	13					
F11	DESTN BIT 08	14	2.4	-2.05	0.11	mA	1-3
F12	MC BIT 8	15	2.4	-0.85	0.045	mA	1-3
F13	MC BIT 9	16					
F14	MC BIT 10	17					
F15	MC BIT 11	18					
F16	MC BIT 12	19					
F17	MC BIT 13	21	2.4	-0.85	0.045	mA	1-3
F18	MC BIT 14	22	2.4	-1.2	0.04	mA	1-3
F19	MC BIT 15	23	2.4	-1.2	0.04	mA	1-3
F20	EN T MODE	35	2.4	-0.6	0.02	mA	1-3
F21	PH N CLK	36	2.4	-4.6	0.12	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
F22	CTRL FRMT	37	2.4	-0.25	0.025	mA	1-3
F23	PH SCE EN	38	2.4	-0.6	0.02	mA	1-3
F24	DEST BIT 4	39	2.4	-0.4	0.02	mA	1-3
F25	DEST BIT 3	41					
F26	DEST BIT 2	42					
F27	DEST BIT 1	43					
F28	DEST BIT 0	44	2.4	-0.4	0.02	mA	1-3
F29	SA BIT 3	45	2.4	-0.6	0.02	mA	1-3
F30	SA BIT 2	46					
F31	SA BIT 1	47					
F32	SA BIT 0	48	2.4	-0.6	0.02	mA	1-3
F33	SR1 BIT 14	50	2.4	-1.2	0.04	mA	1-3
F34	DESTN BIT 07	52	2.4	-2.05	0.11	mA	1-3
F35	DESTN BIT 06	53					
F36	DESTN BIT 05	54					
F37	DESTN BIT 04	55	2.4	-2.05	0.11	mA	1-3
F38	DESTN BIT 03	56	2.4	-2.65	0.13	mA	1-3
F39	DESTN BIT 02	57					
F40	DESTN BIT 01	58					
F41	DESTN BIT 00	59	2.4	-2.65	0.13	mA	1-3
F42	MC BIT 0	70	2.4	-0.2	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
F43	MC BIT 1	71	2.4	-0.2	0.02	mA	1-3
F44	MC BIT 2	72					
F45	MC BIT 3	73					
F46	MC BIT 4	74					
F47	MC BIT 5	76					
F48	MC BIT 6	77					
F49	MC BIT 7	78	2.4	-0.2	0.02	mA	1-3
F50	IR BIT 7	79	2.4	-0.6	0.02	mA	1-3
F51	IR BIT 6	80					
F52	IR BIT 5	82					
F53	IR BIT 4	83					
F54	IR BIT 3	84					
F55	IR BIT 2	85					
F56	IR BIT 1	86					
F57	IR BIT 0	87	2.4	-0.6	0.02	mA	1-3
F58	EMULATE	88	2.4	-0.25	0.025	mA	1-3
F59	ECW BIT 1	89					
F60	ECW BIT 0	91					
F61	TRIG 5	92					
F62	TRIG 4	93					
F63	TRIG 3	94	2.4	-0.25	0.025	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F64	PH B EN	95	2.4	-1.2	0.04	mA	1-3
F65	SB BIT 0	96	2.4	-0.6	0.02	mA	1-3
F66	SB BIT 1	97					
F67	SB BIT 2	98					
F68	SB BIT 3	99					
F69	SB BIT 4	100	2.4	-0.6	0.02	mA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F69
Power supply current limit	2.9 A	NA	F1-F69
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F69
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 F1-F69 F1-F69
Input current measurement	 NA NA	 ± 339.0 uA ± 31.0 uA	 F1 F2,F3

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 108.0 uA	F4-F11
	NA	± 44.8 uA	F12-F17
	NA	± 62.0 uA	F18,F19
	NA	± 31.0 uA	F20
	NA	± 236.0 uA	F21
	NA	± 13.8 uA	F22
	NA	± 31.0 uA	F23
	NA	± 21.0 uA	F24-F28
	NA	± 31.0 uA	F29-F32
	NA	± 62.0 uA	F33
	NA	± 108.0 uA	F34-F37
	NA	± 139.0 uA	F38-F41
	NA	± 11.0 uA	F42-F49
	NA	± 31.0 uA	F50-F57
	NA	± 13.8 uA	F58-F63
	NA	± 62.0 uA	F64
	NA	± 31.0 uA	F65-F69

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	PHASE CLK	2	0.4	-6.6	0.17	mA	1-3
G2	MSTR CLR	3	0.4	-0.6	0.02	mA	1-3
G3	ALU EXT IN	4	0.4	-0.6	0.02	mA	1-3
G4	DESTN BIT 15	6	0.4	-2.05	0.11	mA	1-3
G5	DESTN BIT 14	7					
G6	DESTN BIT 13	8					
G7	DESTN BIT 12	9					
G8	DESTN BIT 11	11					
G9	DESTN BIT 10	12					
G10	DESTN BIT 09	13					
G11	DESTN BIT 08	14	0.4	-2.05	0.11	mA	1-3
G12	MC BIT 8	15	0.4	-0.85	0.045	mA	1-3
G13	MC BIT 9	16					
G14	MC BIT 10	17					
G15	MC BIT 11	18					
G16	MC BIT 12	19					
G17	MC BIT 13	21	0.4	-0.85	0.045	mA	1-3
G18	MC BIT 14	22	0.4	-1.2	0.04	mA	1-3
G19	MC BIT 15	23	0.4	-1.2	0.04	mA	1-3
G20	EN T MODE	35	0.4	-0.6	0.02	mA	1-3
G21	PH N CLK	36	0.4	-4.6	0.12	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G22	CTRL FRMT	37	0.4	-0.25	0.025	mA	1-3
G23	PH SCE EN	38	0.4	-0.6	0.02	mA	1-3
G24	DEST BIT 4	39	0.4	-0.4	0.02	mA	1-3
G25	DEST BIT 3	41					
G26	DEST BIT 2	42					
G27	DEST BIT 1	43					
G28	DEST BIT 0	44	0.4	-0.4	0.02	mA	1-3
G29	SA BIT 3	45	0.4	-0.6	0.02	mA	1-3
G30	SA BIT 2	46					
G31	SA BIT 1	47					
G32	SA BIT 0	48	0.4	-0.6	0.02	mA	1-3
G33	SR1 BIT 14	50	0.4	-1.2	0.04	mA	1-3
G34	DESTN BIT 07	52	0.4	-2.05	0.11	mA	1-3
G35	DESTN BIT 06	53					
G36	DESTN BIT 05	54					
G37	DESTN BIT 04	55	0.4	-2.05	0.11	mA	1-3
G38	DESTN BIT 03	56	0.4	-2.65	0.13	mA	1-3
G39	DESTN BIT 02	57					
G40	DESTN BIT 01	58					
G41	DESTN BIT 00	59	0.4	-2.65	0.13	mA	1-3
G42	MC BIT 0	70	0.4	-0.2	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G43	MC BIT 1	71	0.4	-0.2	0.02	mA	1-3
G44	MC BIT 2	72					
G45	MC BIT 3	73					
G46	MC BIT 4	74					
G47	MC BIT 5	76					
G48	MC BIT 6	77					
G49	MC BIT 7	78	0.4	-0.2	0.02	mA	1-3
G50	IR BIT 7	79	0.4	-0.6	0.02	mA	1-3
G51	IR BIT 6	80					
G52	IR BIT 5	82					
G53	IR BIT 4	83					
G54	IR BIT 3	84					
G55	IR BIT 2	85					
G56	IR BIT 1	86					
G57	IR BIT 0	87	0.4	-0.6	0.02	mA	1-3
G58	EMULATE	88	0.4	-0.25	0.025	mA	1-3
G59	ECW BIT 1	89					
G60	ECW BIT 0	91					
G61	TRIG 5	92					
G62	TRIG 4	93					
G63	TRIG 3	94	0.4	-0.25	0.025	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G64	PH B EN	95	0.4	-1.2	0.04	mA	1-3
G65	SB BIT 0	96	0.4	-0.6	0.02	mA	1-3
G66	SB BIT 1	97					
G67	SB BIT 2	98					
G68	SB BIT 3	99					
G69	SB BIT 4	100	0.4	-0.6	0.02	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G69
Power supply current limit	2.9 A	NA	G1-G69
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G69
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	G1-G69 G1-G69
Input current measurement	NA NA	± 339.0 uA ± 31.0 uA	G1 G2,G3

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 108.0 uA	G4-G11
	NA	± 44.8 uA	G12-G17
	NA	± 62.0 uA	G18,G19
	NA	± 31.0 uA	G20
	NA	± 236.0 uA	G21
	NA	± 13.8 uA	G22
	NA	± 31.0 uA	G23
	NA	± 21.0 uA	G24-G28
	NA	± 31.0 uA	G29-G32
	NA	± 62.0 uA	G33
	NA	± 108.0 uA	G34-G37
	NA	± 139.0 uA	G38-G41
	NA	± 11.0 uA	G42-G49
	NA	± 31.0 uA	G50-G57
	NA	± 13.8 uA	G58-G63
	NA	± 62.0 uA	G64
	NA	± 31.0 uA	G65-G69

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	ALU FUNCT	5	-0.4	2.4	4.5	V	1-4
H2	DESTN BIT 15	6					
H3	DESTN BIT 14	7					
H4	DESTN BIT 13	8					
H5	DESTN BIT 12	9					
H6	DESTN BIT 11	11					
H7	DESTN BIT 10	12					
H8	DESTN BIT 09	13					
H9	DESTN BIT 08	14	-0.4	2.4	4.5	V	1-4
H10	0 FILL	24	-0.16	2.4	4.5	V	1-4
H11	0 FILL	24					
H12	0 FILL	24					
H13	GR BIT 15	26					
H14	GR BIT 15	26					
H15	GR BIT 15	26					
H16	GR BIT 15	26					
H17	GR BIT 14	27					
H18	GR BIT 14	27					
H19	GR BIT 14	27					
H20	GR BIT 14	27					
H21	GR BIT 13	28	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H22	GR BIT 13	28	-0.16	2.4	4.5	V	1-4
H23	GR BIT 13	28					
H24	GR BIT 13	28					
H25	GR BIT 12	29					
H26	GR BIT 12	29					
H27	GR BIT 12	29					
H28	GR BIT 12	29					
H29	GR BIT 11	30					
H30	GR BIT 11	30					
H31	GR BIT 11	30					
H32	GR BIT 11	30					
H33	GR BIT 10	32					
H34	GR BIT 10	32					
H35	GR BIT 10	32					
H36	GR BIT 10	32					
H37	GR BIT 09	33					
H38	GR BIT 09	33					
H39	GR BIT 09	33					
H40	GR BIT 09	33	-0.16	2.4	4.5	V	1-4
H41	SER DATA OUT	34	-0.4	2.4	4.5	V	1-4
H42	SGR NE M	49	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H43	DESTN BIT 07	52	-0.4	2.4	4.5	V	1-4
H44	DESTN BIT 06	53					
H45	DESTN BIT 05	54					
H46	DESTN BIT 04	55					
H47	DESTN BIT 03	56					
H48	DESTN BIT 02	57					
H49	DESTN BIT 01	58					
H50	DESTN BIT 00	59	-0.4	2.4	4.5	V	1-4
H51	UP BYTE SEL	61	-0.16	2.4	4.5	V	1-4
H52	UP BYTE SEL	61					
H53	GR BIT 03	62					
H54	GR BIT 03	62					
H55	GR BIT 03	62					
H56	GR BIT 03	62					
H57	GR BIT 02	63					
H58	GR BIT 02	63					
H59	GR BIT 02	63					
H60	GR BIT 02	63					
H61	GR BIT 01	64					
H62	GR BIT 01	64					
H63	GR BIT 01	64	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H64	GR BIT 01	64	-0.16	2.4	4.5	V	1-4
H65	GR BIT 00	65					
H66	GR BIT 00	65					
H67	GR BIT 00	65					
H68	GR BIT 00	65					
H69	GR BIT 07	66					
H70	GR BIT 07	66					
H71	GR BIT 07	66					
H72	GR BIT 07	66					
H73	GR BIT 06	67					
H74	GR BIT 06	67					
H75	GR BIT 06	67					
H76	GR BIT 06	67					
H77	GR BIT 05	68					
H78	GR BIT 05	68					
H79	GR BIT 05	68					
H80	GR BIT 05	68					
H81	GR BIT 04	69					
H82	GR BIT 04	69					
H83	GR BIT 04	69					
H84	GR BIT 04	69	-0.16	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H84
Power supply current limit	2.9 A	NA	H1-H84
Forcing current, (Ioh)	-0.4 mA -0.16 mA -0.4 mA -0.16 mA -0.4 mA -0.16 mA	± 40 uA ± 16 uA ± 40 uA ± 16 uA ± 40 uA ± 16 uA	H1-H9 H10-H40 H41 H42 H43-H50 H51-H84
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 H1-H84 H1-H84
Output voltage measurement	NA	± 105 mV	H1-H84

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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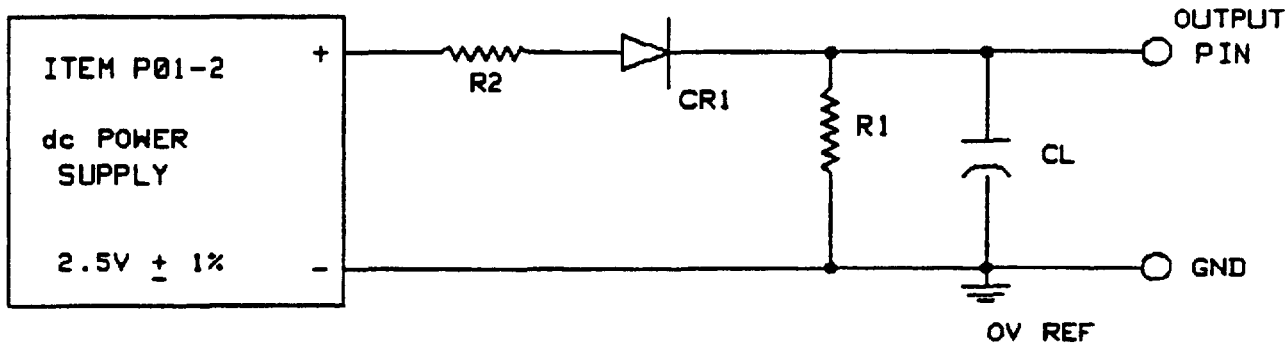
NOTES: - Continued.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R2 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R1 for each output load shall be as specified in the table below.

Pin number	R1 1/ (ohms)	Iol (mA)	Vol (volts)
5,6,7,8,9,11,12, 13,14,34,52,53, 54,55,56,57,58, 59	6200	6.0	0.5
24,26,27,28,29, 30,32,33,49,61, 62,63,64,65,66, 67,68,69	16000	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	ALU FUNCT	5	6.0	0.0	0.5	V	1-4
J2	DESTN BIT 15	6					
J3	DESTN BIT 14	7					
J4	DESTN BIT 13	8					
J5	DESTN BIT 12	9					
J6	DESTN BIT 11	11					
J7	DESTN BIT 10	12					
J8	DESTN BIT 09	13					
J9	DESTN BIT 08	14	6.0	0.0	0.5	V	1-4
J10	0 FILL	24	1.6	0.0	0.5	V	1-4
J11	0 FILL	24					
J12	0 FILL	24					
J13	0 FILL	24					
J14	GR BIT 15	26					
J15	GR BIT 15	26					
J16	GR BIT 15	26					
J17	GR BIT 15	26					
J18	GR BIT 14	27					
J19	GR BIT 14	27					
J20	GR BIT 14	27					
J21	GR BIT 14	27	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J22	GR BIT 13	28	1.6	0.0	0.5	V	1-4
J23	GR BIT 13	28					
J24	GR BIT 13	28					
J25	GR BIT 13	28					
J26	GR BIT 12	29					
J27	GR BIT 12	29					
J28	GR BIT 12	29					
J29	GR BIT 12	29					
J30	GR BIT 11	30					
J31	GR BIT 11	30					
J32	GR BIT 11	30					
J33	GR BIT 11	30					
J34	GR BIT 10	32					
J35	GR BIT 10	32					
J36	GR BIT 10	32					
J37	GR BIT 10	32					
J38	GR BIT 09	33					
J39	GR BIT 09	33					
J40	GR BIT 09	33					
J41	GR BIT 09	33	1.6	0.0	0.5	V	1-4
J42	SER DATA OUT	34	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J43	SGR NE M	49	1.6	0.0	0.5	V	1-4
J44	DESTN BIT 07	52	6.0	0.0	0.5	V	1-4
J45	DESTN BIT 06	53					
J46	DESTN BIT 05	54					
J47	DESTN BIT 04	55					
J48	DESTN BIT 03	56					
J49	DESTN BIT 02	57					
J50	DESTN BIT 01	58					
J51	DESTN BIT 00	59	6.0	0.0	0.5	V	1-4
J52	UP BYTE SEL	61	1.6	0.0	0.5	V	1-4
J53	UP BYTE SEL	61					
J54	GR BIT 03	62					
J55	GR BIT 03	62					
J56	GR BIT 03	62					
J57	GR BIT 03	62					
J58	GR BIT 02	63					
J59	GR BIT 02	63					
J60	GR BIT 02	63					
J61	GR BIT 02	63					
J62	GR BIT 01	64					
J63	GR BIT 01	64	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J64	GR BIT 01	64	1.6	0.0	0.5	V	1-4
J65	GR BIT 01	64					
J66	GR BIT 00	65					
J67	GR BIT 00	65					
J68	GR BIT 00	65					
J69	GR BIT 00	65					
J70	GR BIT 07	66					
J71	GR BIT 07	66					
J72	GR BIT 07	66					
J73	GR BIT 07	66					
J74	GR BIT 06	67					
J75	GR BIT 06	67					
J76	GR BIT 06	67					
J77	GR BIT 06	67					
J78	GR BIT 05	68					
J79	GR BIT 05	68					
J80	GR BIT 05	68					
J81	GR BIT 05	68					
J82	GR BIT 04	69					
J83	GR BIT 04	69					
J84	GR BIT 04	69					
J85	GR BIT 04	69	1.6	0.0	0.5	V	1-4

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J85
Power supply current limit	2.9 A	NA	J1-J85
Forcing current, (Ioh)	6.0 mA 1.6 mA 6.0 mA 1.6 mA 6.0 mA 1.6 mA	± 600 uA ± 160 uA ± 600 uA ± 160 uA ± 600 uA ± 160 uA	J1-J9 J10-J41 J42 J43 J44-J51 J52-J85
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J85 J1-J85
Output voltage measurement	NA	± 25 mV	J1-J85

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
High level							
K1	0 FILL	24	2.4	-0.17	0.17	mA	1-3
K2	GR BIT 15	26					
K3	GR BIT 14	27					
K4	GR BIT 13	28					
K5	GR BIT 12	29					
K6	GR BIT 11	30					
K7	GR BIT 10	32					
K8	GR BIT 09	33					
K9	GR BIT 03	62					
K10	GR BIT 02	63					
K11	GR BIT 01	64					
K12	GR BIT 00	65					
K13	GR BIT 07	66					
K14	GR BIT 06	67					
K15	GR BIT 05	68					
K16	GR BIT 04	69	2.4	-0.17	0.17	mA	1-3
Low level							
K17	0 FILL	24	0.4	-0.17	0.17	mA	1-3
K18	GR BIT 15	26					
K19	GR BIT 14	27	0.4	-0.17	0.17	mA	1-3

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TABLE XI. Output leakage current, high and low level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K20	GR BIT 13	28	0.4	-0.17	0.17	mA	1-3
K21	GR BIT 12	29					
K22	GR BIT 11	30					
K23	GR BIT 10	32					
K24	GR BIT 09	33					
K25	GR BIT 03	62					
K26	GR BIT 02	63					
K27	GR BIT 01	64					
K28	GR BIT 00	65					
K29	GR BIT 07	66					
K30	GR BIT 06	67					
K31	GR BIT 05	68					
K32	GR BIT 04	69	0.4	-0.17	0.17	mA	1-3

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NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K32
Power supply current limit	2.9 A	NA	K1-K32
Forcing voltage, (Voh) (Vol)	 2.4 V 0.4 V	 ± 50 mV ± 50 mV	 K1-K16 K17-K32
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 K1-K32 K1-K32
Output current measurement	NA	± 17 uA	K1-K32

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	PHASE CLK	2	-18.0	-1.2	0.0	V	1-3
L2	MSTR CLR	3					
L3	ALU EXT IN	4					
L4	MC BIT 8	15					
L5	MC BIT 9	16					
L6	MC BIT 10	17					
L7	MC BIT 11	18					
L8	MC BIT 12	19					
L9	MC BIT 13	21					
L10	MC BIT 14	22					
L11	MC BIT 15	23					
L12	EN T MODE	35					
L13	PH N CLK	36	-18.0	-1.2	0.0	V	1-3
L14	CTRL FRMT	37	-18.0	-1.5	0.0	V	1-3
L15	PH SCE EN	38	-18.0	-1.2	0.0	V	1-3
L16	DEST BIT 4	39	-18.0	-1.5	0.0	V	1-3
L17	DEST BIT 3	41					
L18	DEST BIT 2	42					
L19	DEST BIT 1	43					
L20	DEST BIT 0	44	-18.0	-1.5	0.0	V	1-3
L21	SA BIT 3	45	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L22	SA BIT 2	46	-18.0	-1.2	0.0	V	1-3
L23	SA BIT 1	47					
L24	SA BIT 0	48					
L25	SR1 BIT 14	50	-18.0	-1.2	0.0	V	1-3
L26	MC BIT 0	70	-18.0	-1.5	0.0	V	1-3
L27	MC BIT 1	71					
L28	MC BIT 2	72					
L29	MC BIT 3	73					
L30	MC BIT 4	74					
L31	MC BIT 5	76					
L32	MC BIT 6	77					
L33	MC BIT 7	78	-18.0	-1.5	0.0	V	1-3
L34	IR BIT 7	79	-18.0	-1.2	0.0	V	1-3
L35	IR BIT 6	80					
L36	IR BIT 5	82					
L37	IR BIT 4	83					
L38	IR BIT 3	84					
L39	IR BIT 2	85					
L40	IR BIT 1	86					
L41	IR BIT 0	87	-18.0	-1.2	0.0	V	1-3
L42	EMULATE	88	-18.0	-1.5	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L43	ECW BIT 1	89	-18.0	-1.5	0.0	V	1-3
L44	ECW BIT 0	91					
L45	TRIG 5	92					
L46	TRIG 4	93					
L47	TRIG 3	94	-18.0	-1.5	0.0	V	1-3
L48	PH B EN	95	-18.0	-1.2	0.0	V	1-3
L49	SB BIT 0	96					
L50	SB BIT 1	97					
L51	SB BIT 2	98					
L52	SB BIT 3	99					
L53	SB BIT 4	100	-18.0	-1.2	0.0	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L53
Power supply current limit	NA	NA	L1-L53
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L53

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage			
high, (VH)	NA	NA	L1-L53
low, (VL)	NA	NA	L1-L53
Input voltage measurement	NA	± 60 mV	L1-L13
	NA	± 75 mV	L14
	NA	± 60 mV	L15
	NA	± 75 mV	L16-L20
	NA	± 60 mV	L21-L25
	NA	± 75 mV	L26-L33
	NA	± 60 mV	L34-L41
	NA	± 75 mV	L42-L47
	NA	± 60 mV	L48-L53

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	4.9	9.1	uF	1-4
M2	PSFC	51	6.3	11.7	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 210 nF ± 270 nF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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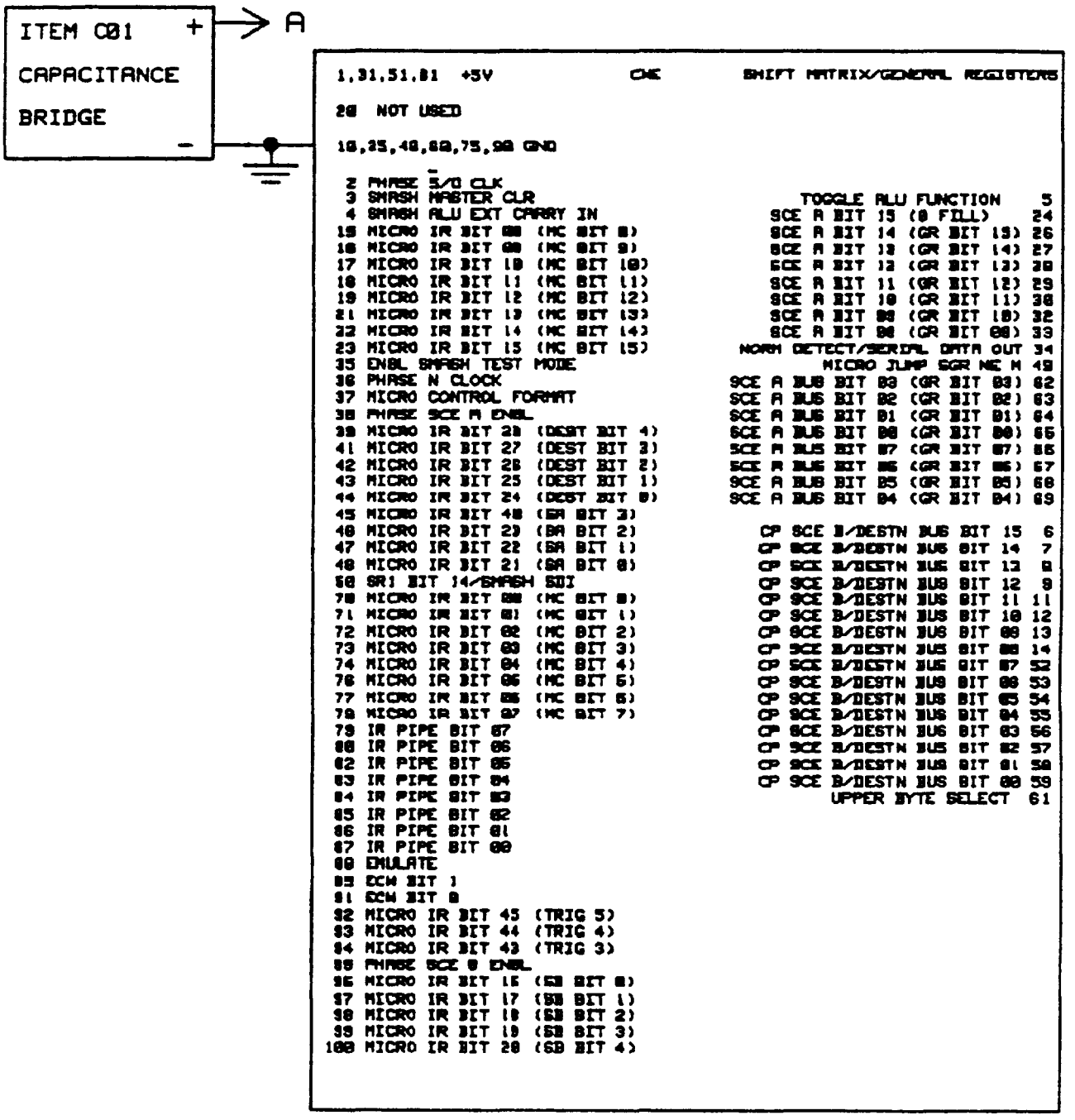


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31, 51,81	800	1920	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	2.9 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 56 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes	Figs
P1	Functional test minimum power supply voltage maximum pattern rate (250 nS)	1-3	9,11
P2	Functional test maximum power supply voltage maximum pattern rate (250 nS)	1-3	9,11

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	2.9 A 2.9 A	NA NA	P1 P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 25 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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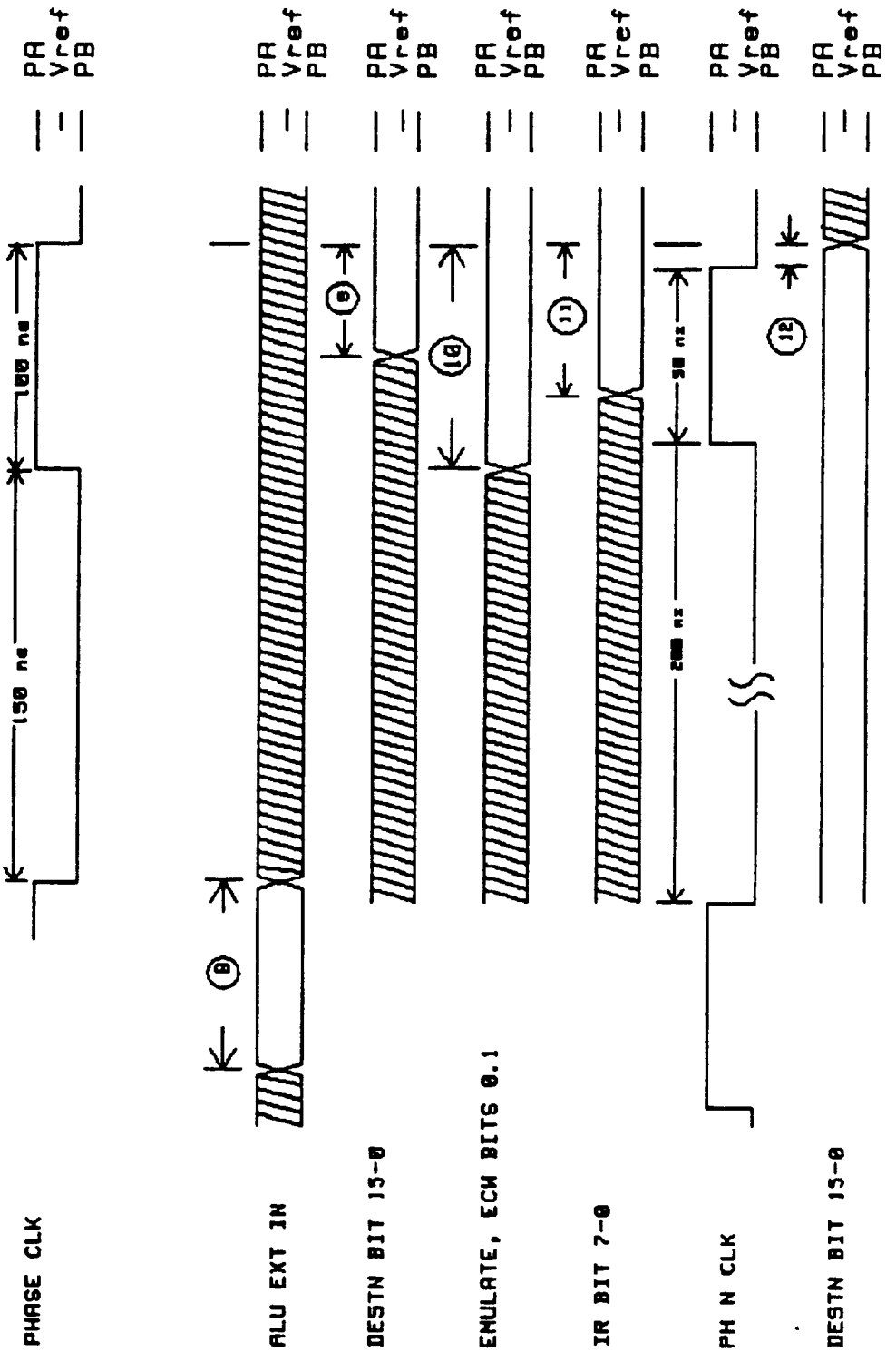


FIGURE 11. Input timing parameters for table XV.

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NOTES:

1. (8) ALU EXT IN to PHASE CLK. Setup time: minimum 60 nS.

2. (9) DESTN BIT 15-00 to PHASE CLK. Setup time: minimum 52 nS.

3. (10) EMULATE to PHASE CLK, ECW BIT 0 to PHASE CLK, ECW BIT 1 to PHASE CLK. Setup time: minimum 87 nS.

4. (11) IR BIT 7-00 to PHASE CLK. Setup time: minimum 46 nS.

5. (12) PH N CLK to DESTN BIT 15-00. Hold time: minimum 17 nS.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (250 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (250 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (250 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (250 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	2.9 A 2.9 A	Q1,Q3 Q2,Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 NA NA	 Q1-Q4 Q1-Q4
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 100 mV ± 100 mV ± 100 mV	 NA NA NA	 Q1,Q3 Q2,Q4 Q1-Q4

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

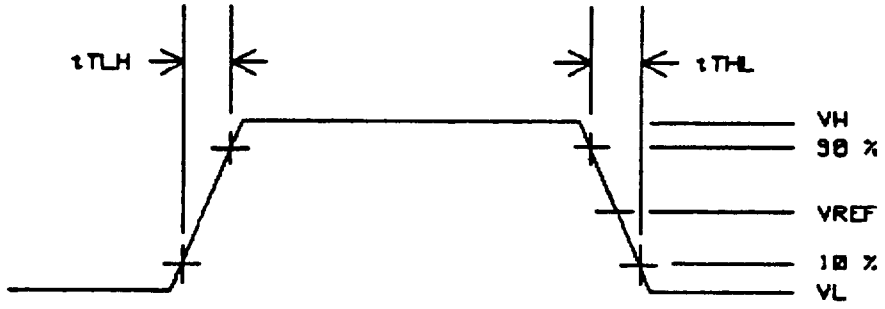
(b) The output compare levels shall be as follows:

High level: Vref $\leq$ Voh $\leq$ Vcc
Low level: 0.0 V $\leq$ Vol $\leq$ Vref

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. The test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	PHASE CLK to ALU FUNCT	-	-	-	129	129	129	nS	12
2	tpd	PH B EN to DESTN BIT 00	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 01	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 02	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 03	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 04	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 05	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 06	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 07	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 08	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 09	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 10	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 11	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 12	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 13	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 14	-	-	-	85	85	85	nS	12
2	tpd	PH B EN to DESTN BIT 15	-	-	-	85	85	85	nS	12
3	tpd	PHASE CLK to DESTN BIT 00	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 10	-	-	-	140	140	140	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tpd	PHASE CLK to DESTN BIT 02	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 03	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 04	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 05	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 06	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 07	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 08	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 09	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 10	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 11	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 12	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 13	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 14	-	-	-	140	140	140	nS	12
3	tpd	PHASE CLK to DESTN BIT 15	-	-	-	140	140	140	nS	12
4	tpd	PH SCE EN to GR BIT 00	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 01	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 02	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 03	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 04	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 05	-	-	-	84	84	84	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	PH SCE EN to GR BIT 06	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 07	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to 0 FILL	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 09	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 10	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 11	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT11,12	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 13	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 14	-	-	-	84	84	84	nS	12
4	tpd	PH SCE EN to GR BIT 15	-	-	-	84	84	84	nS	12
5	tpd	SR1 BIT 14 to 0 FILL	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 15	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 14	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 13	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 12	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 11	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 10	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 09	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 07	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 06	-	-	-	76	76	76	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tpd	SR1 BIT 14 to GR BIT 05	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 04	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 03	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 02	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 01	-	-	-	76	76	76	nS	12
5	tpd	SR1 BIT 14 to GR BIT 00	-	-	-	76	76	76	nS	12
6	tpd	PHASE CLK to UP BYTE SEL	-	-	-	96	96	96	nS	12
7	tpd	PHASE CLK to SGR NE M	-	-	-	84	84	84	nS	12
8	tsu	ALU EXT IN to PHASE CLK	60	60	60	-	-	-	nS	12
9	tsu	DESTN BIT 15 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 14 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 13 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 12 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 11 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 10 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 09 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 08 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 07 to PHASE CLK	52	52	52	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
9	tsu	DESTN BIT 06 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 05 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 04 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 03 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 02 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 01 to PHASE CLK	52	52	52	-	-	-	nS	12
9	tsu	DESTN BIT 00 to PHASE CLK	52	52	52	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 15	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 14	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 13	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 12	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 11	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 10	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 09	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 08	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 07	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 06	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 05	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 04	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 03	17	17	17	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
10	th	PH N CLK to DESTN BIT 02	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 01	17	17	17	-	-	-	nS	12
10	th	PH N CLK to DESTN BIT 00	17	17	17	-	-	-	nS	12
11	tsu	EMULATE to PHASE CLK	87	87	87	-	-	-	nS	12
11	tsu	ECW BIT 1 to PHASE CLK	87	87	87	-	-	-	nS	12
11	tsu	ECW BIT 0 to PHASE CLK	87	87	87	-	-	-	nS	12
12	tsu	IR BIT 7 to PHASE CLK	46	46	46	-	-	-	nS	12
12	tsu	IR BIT 6 to PHASE CLK	46	46	46	-	-	-	nS	12
12	tsu	IR BIT 5 to PHASE CLK	46	46	46	-	-	-	nS	12
12	tsu	IR BIT 4 to PHASE CLK	46	46	46	-	-	-	nS	12
12	tsu	IR BIT 3 to PHASE CLK	46	46	46	-	-	-	nS	12
12	tsu	IR BIT 2 to PHASE CLK	46	46	46	-	-	-	nS	12
12	tsu	IR BIT 1 to PHASE CLK	46	46	46	-	-	-	nS	12
12	tsu	IR BIT 0 to PHASE CLK	46	46	46	-	-	-	nS	12

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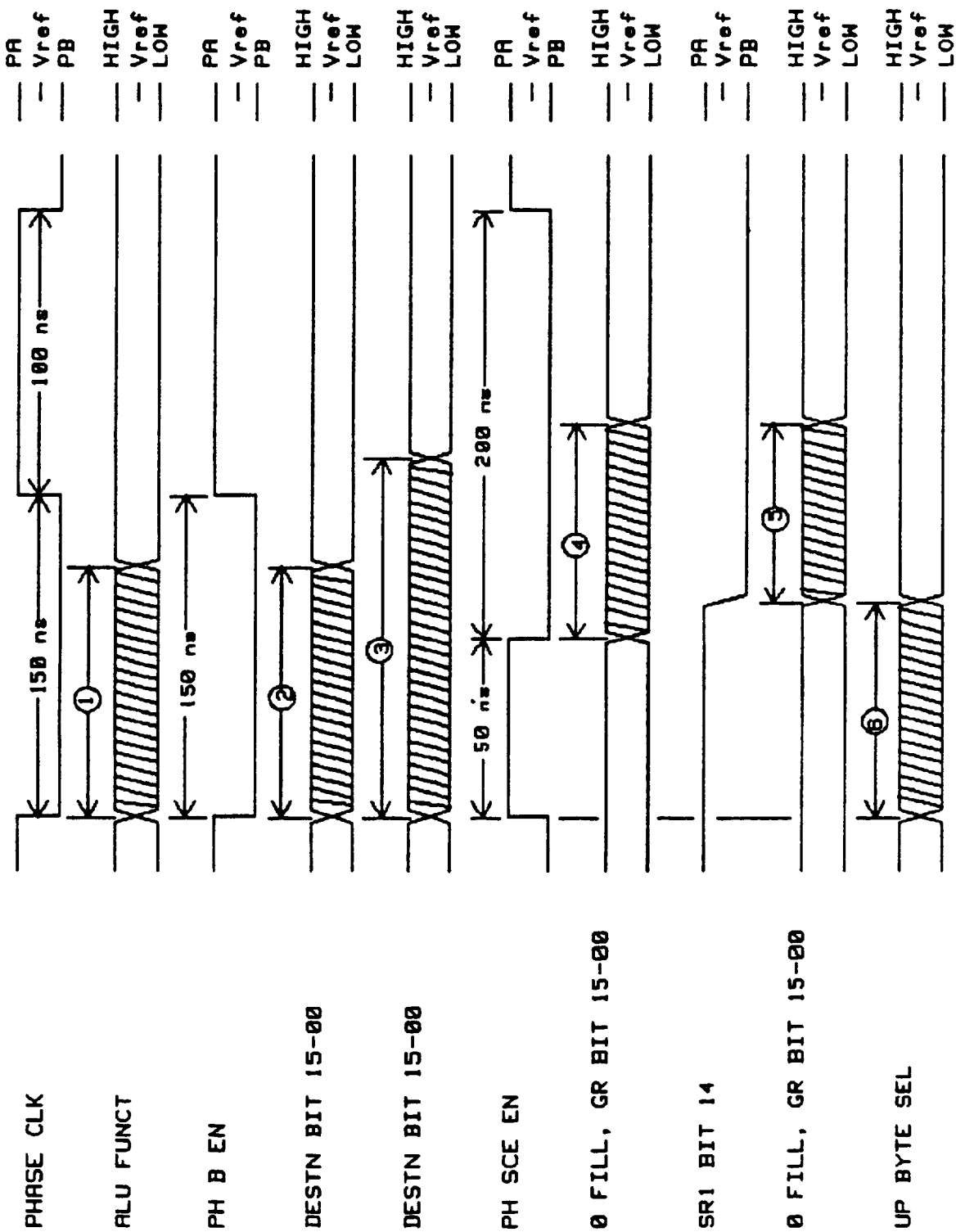


FIGURE 12a. Dynamic waveforms for table XVII.

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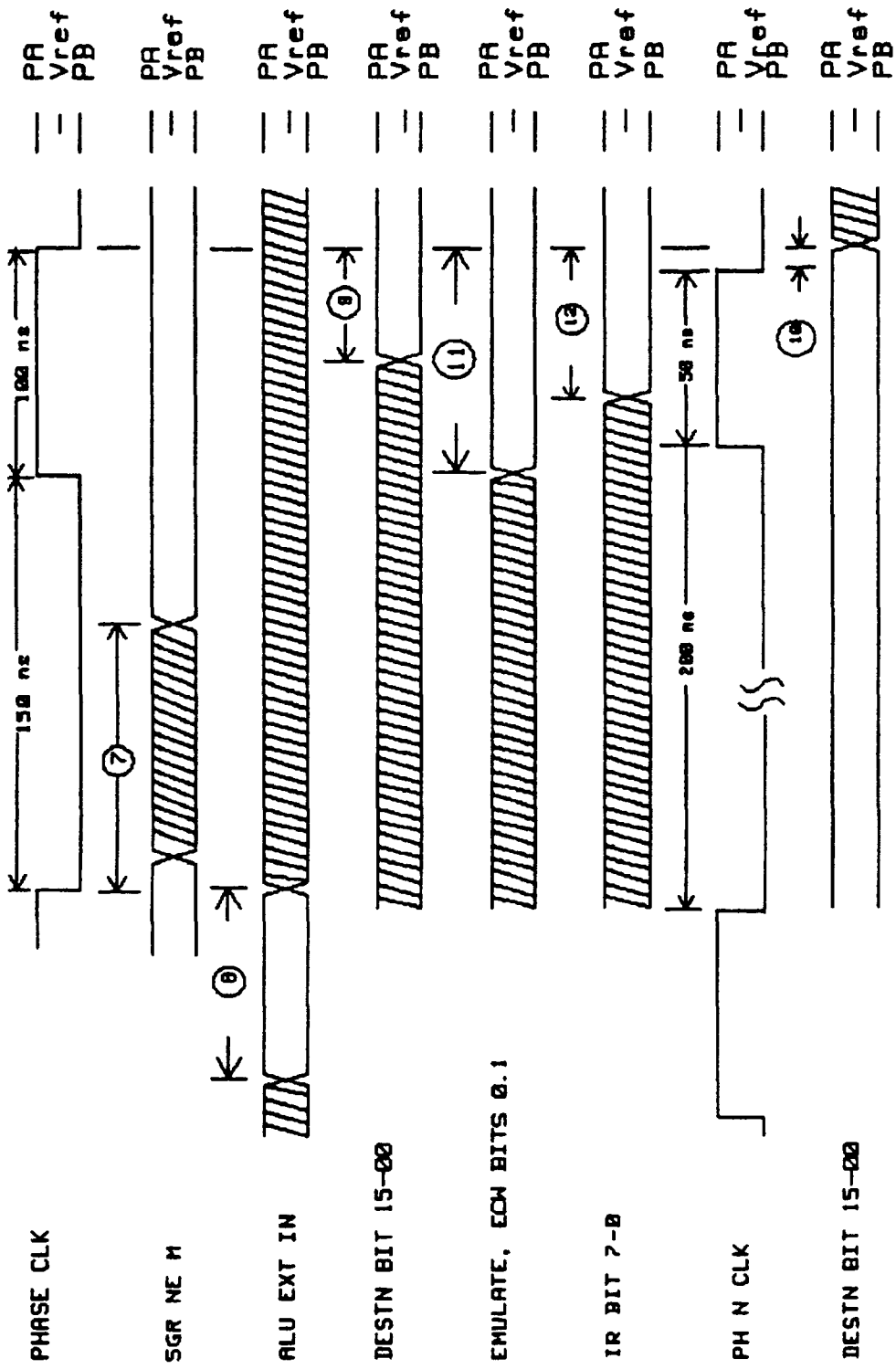


FIGURE 12b. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	PHASE CLK	2	8	125	pF	1-4
R2	MSTR CLR	3	2	75	pF	1-4
R3	ALU EXT IN	4	2	75	pF	1-4
R4	MC BIT 8	15	4	100	pF	1-4
R5	MC BIT 9	16				
R6	MC BIT 10	17				
R7	MC BIT 11	18				
R8	MC BIT 12	19				
R9	MC BIT 13	21				
R10	MC BIT 14	22				
R11	MC BIT 15	23	4	100	pF	1-4
R12	EN T MODE	35	2	50	pF	1-4
R13	PH N CLK	36	6	100	pF	1-4
R14	CTRL FRMT	37	2	75	pF	1-4
R15	PH SCE EN	38	2	50	pF	1-4
R16	DEST BIT 4	39	2	75	pF	1-4
R17	DEST BIT 3	41				
R18	DEST BIT 2	42	2	75	pF	1-4
R19	DEST BIT 1	43	2	50	pF	1-4
R20	DEST BIT 0	44				
R21	SA BIT 3	45	2	50	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
R22	SA BIT 2	46	2	50	pF	1-4
R23	SA BIT 1	47				
R24	SA BIT 0	48	2	50	pF	1-4
R25	SR1 BIT 14	50	4	75	pF	1-4
R26	MC BIT 0	70	2	50	pF	1-4
R27	MC BIT 1	71				
R28	MC BIT 2	72				
R29	MC BIT 3	73				
R30	MC BIT 4	74				
R31	MC BIT 5	76				
R32	MC BIT 6	77				
R33	MC BIT 7	78				
R34	IR BIT 7	79				
R35	IR BIT 6	80				
R36	IR BIT 5	82				
R37	IR BIT 4	83				
R38	IR BIT 3	84				
R39	IR BIT 2	85				
R40	IR BIT 1	86				
R41	IR BIT 0	87				
R42	EMULATE	88	2	50	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R43	ECW BIT 1	89	2	50	pF	1-4
R44	ECW BIT 0	91				
R45	TRIG 5	92				
R46	TRIG 4	93				
R47	TRIG 3	94	2	50	pF	1-4
R48	PH B EN	95	4	75	pF	1-4
R49	SB BIT 0	96	2	50	pF	1-4
R50	SB BIT 1	97				
R51	SB BIT 2	98				
R52	SB BIT 3	99				
R53	SB BIT 4	100	2	50	pF	1-4
Bidirectionals						
R54	DESTN BIT 15	6	10	150	pF	1-4
R55	DESTN BIT 14	7	10	175	pF	1-4
R56	DESTN BIT 13	8	10	175	pF	1-4
R57	DESTN BIT 12	9	10	150	pF	1-4
R58	DESTN BIT 11	11	10	125	pF	1-4
R59	DESTN BIT 10	12	10	150	pF	1-4
R60	DESTN BIT 09	13				
R61	DESTN BIT 08	14				
R62	DESTN BIT 07	52				
R63	DESTN BIT 06	53	10	150	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R64	DESTN BIT 05	54	10	150	pF	1-4
R65	DESTN BIT 04	55	10	125	pF	1-4
R66	DESTN BIT 03	56	12	175	pF	1-4
R67	DESTN BIT 02	57				
R68	DESTN BIT 01	58	12	175	pF	1-4
R69	DESTN BIT 00	59	12	150	pF	1-4
R70	UP BYTE SEL	61	6	100	pF	1-4
Outputs						
R71	0 FILL	24	8	150	pF	1-4
R72	GR BIT 15	26				
R73	GR BIT 14	27				
R74	GR BIT 13	28	8	150	pF	1-4
R75	GR BIT 12	29	8	125	pF	1-4
R76	GR BIT 11	30				
R77	GR BIT 10	32				
R78	GR BIT 09	33				
R79	GR BIT 03	62				
R80	GR BIT 02	63				
R81	GR BIT 01	64				
R82	GR BIT 00	65				
R83	GR BIT 07	66				
R84	GR BIT 06	67	8	125	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
R85	GR BIT 05	68	8	125	pF	1-4
R86	GR BIT 04	69	8	125	pF	1-4

NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R86
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R86
Capacitance measurement	NA	± 5.85 pF	R1
	NA	± 3.65 pF	R2,R3
	NA	± 4.8 pF	R4-R11
	NA	± 2.4 pF	R12
	NA	± 4.7 pF	R13
	NA	± 3.65 pF	R14
	NA	± 2.4 pF	R15
	NA	± 3.65 pF	R16-R18
	NA	± 2.4 pF	R19-R24
	NA	± 3.55 pF	R25
	NA	± 2.4 pF	R26-R47
	NA	± 3.55 pF	R48
	NA	± 2.4 pF	R49-R53
	NA	± 7.0 pF	R54
	NA	± 8.25 pF	R55,R56
	NA	± 7.0 pF	R57
	NA	± 5.75 pF	R58
	NA	± 7.0 pF	R59-R64

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Capacitance measurement	NA	± 5.75 pF	R65
	NA	± 8.15 pF	R66-R68
	NA	± 6.9 pF	R69
	NA	± 4.7 pF	R70
	NA	± 7.1 pF	R71-R74
	NA	± 5.85 pF	R75-R86

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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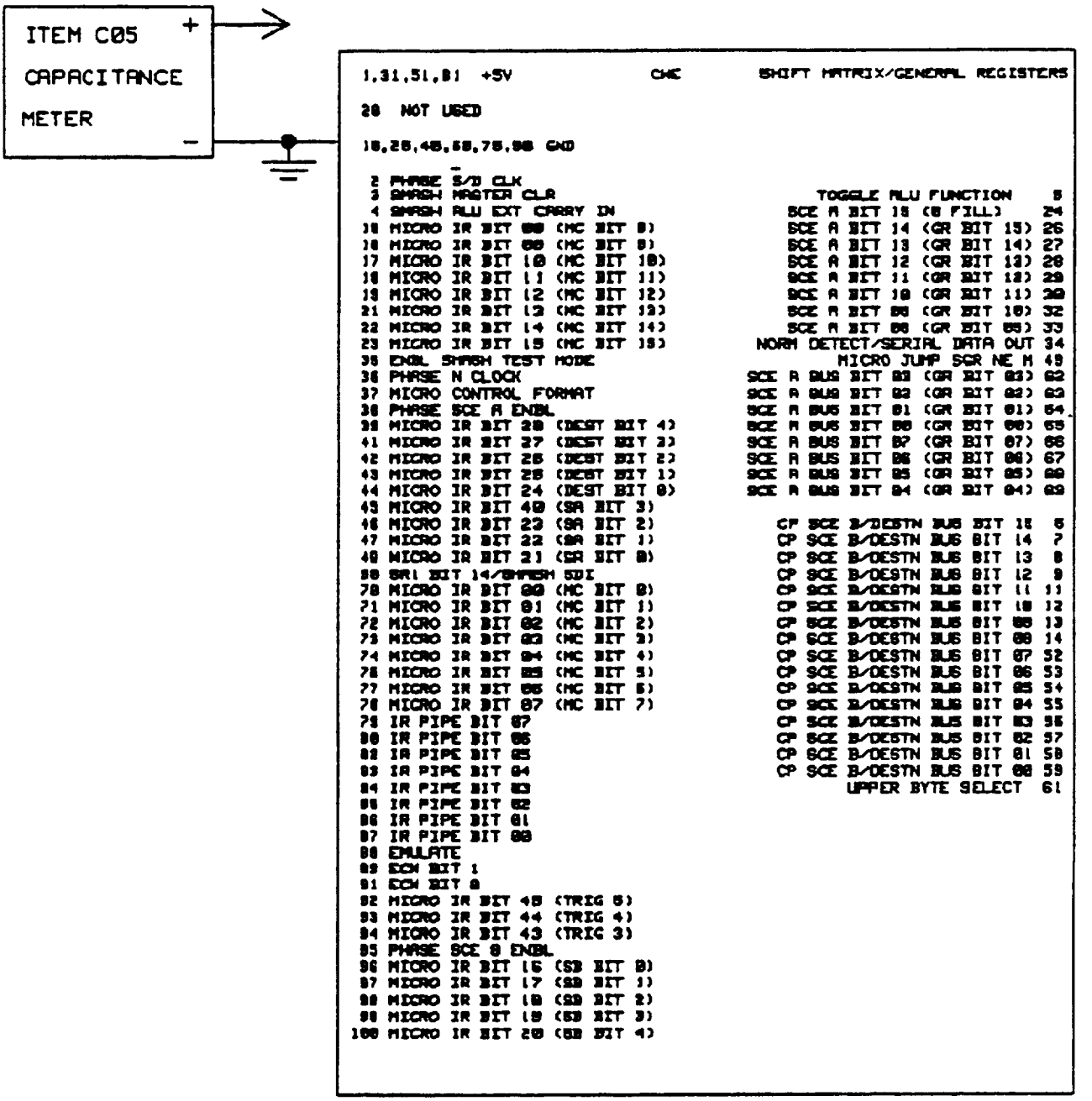


FIGURE 13. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs.
T1	PSTA	1	0	600	mV	1-4	14,15
T2	PSTA	51	0	1000	mV	1-4	14,15

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	2.9 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	 NA NA	 ± 30.0 mV ± 50.0 mV	 T1 T2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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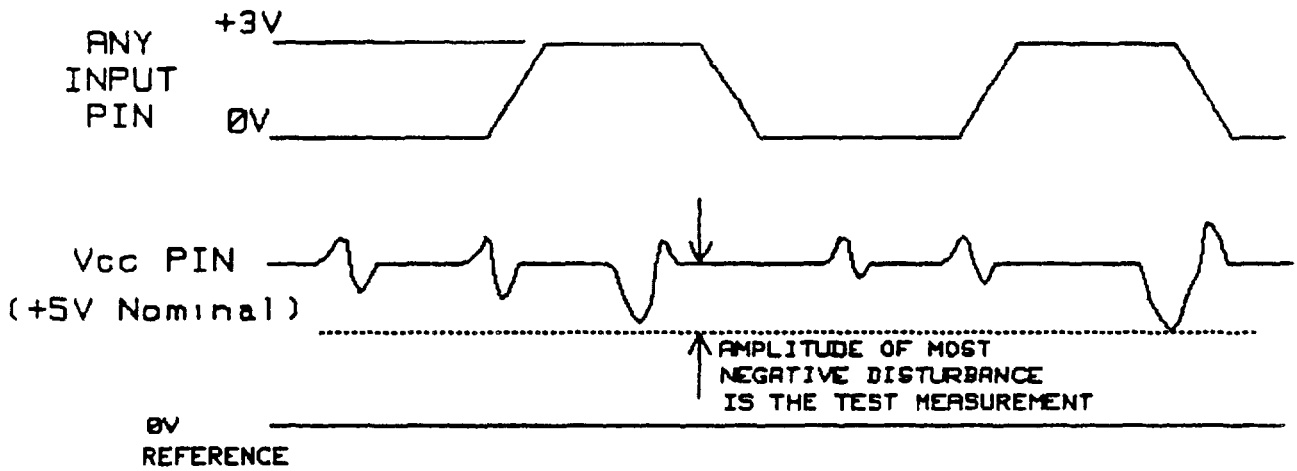
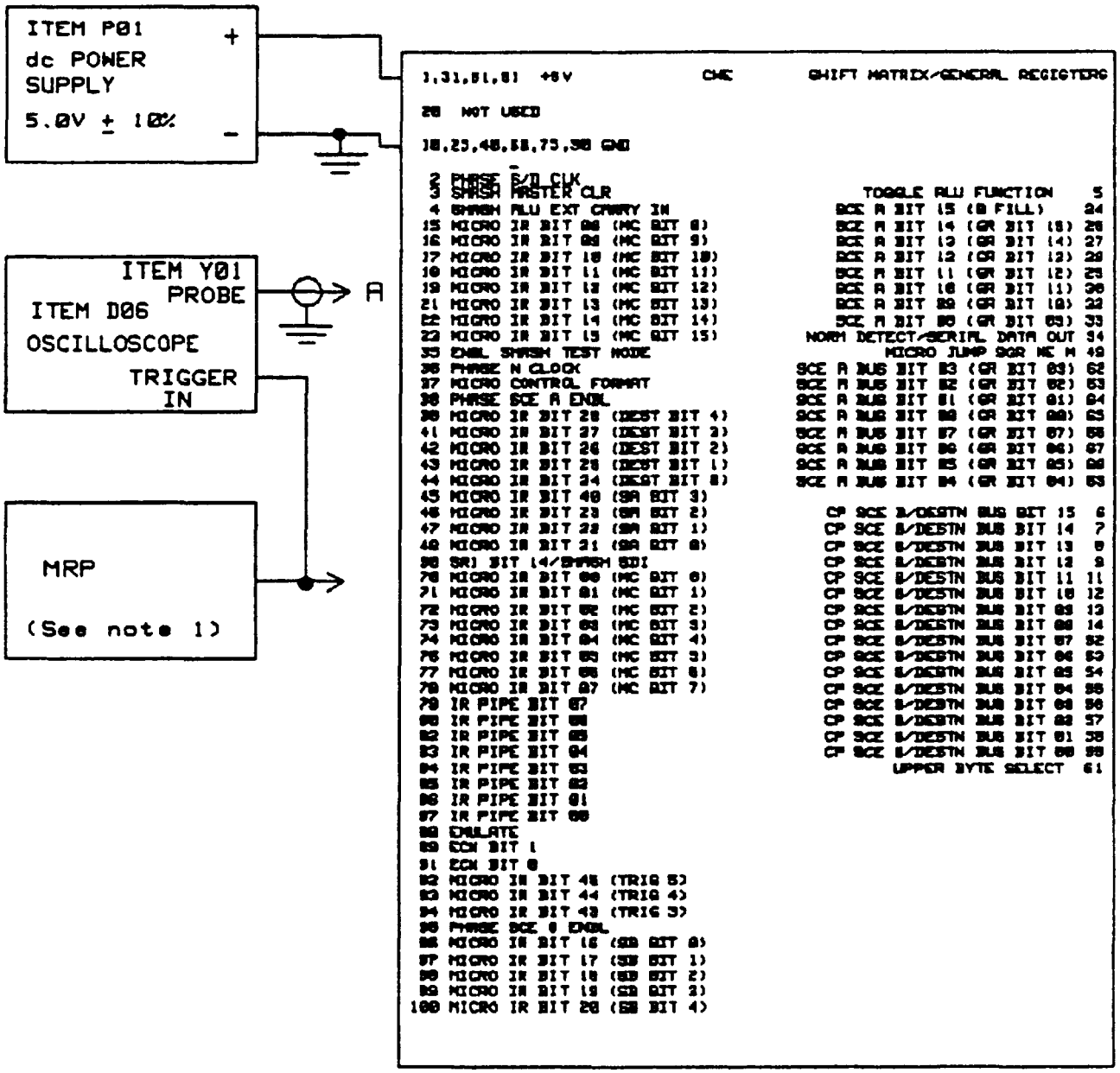


FIGURE 14. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XIX.

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